3.3 Rings and Modules

3.3.1 Commutative Rings

Definition 3.3.1 (Ring)
A ring consists of a triple $(A, +, \cdot)$ where A is a set and $+$ and $\cdot$ are laws of composition ("additive" and "multiplicative" respectively) such that the following holds

- $(A, +)$ *is an* ***abelian group****, whose identity element we refer to as* 0_A*.*
- $(A, \cdot)$ *is a* ***monoid****, whose identity element we refer to as* 1_A
- $+$ *and* $\cdot$ *satisfy the* ***distributive property****, that is for all* $x, y, z \in A$

$$x \cdot (y + z) = x \cdot y + x \cdot z$$

$$(x + y) \cdot z = x \cdot z + y \cdot z$$

For $x \in A$ we write the additive inverse as $-x$, and abbreviate multiplication $x \cdot y =: xy$.

We say that A is a ***zero-ring*** *(or trivial) if $0_A = 1_A \iff A = \{0\}$.*

A is ***commutative*** *if in addition $xy = yx$ i.e. $(A, \cdot)$ is abelian.*

Example 3.3.2
The set of integers $\mathbb{Z}$ (see the Integers subsection) is the canonical example of a ring with operations of addition and multiplication.

Definition 3.3.3 (Subring)
A ***subring*** *of a ring A is a subset B such that*

- $0_A, 1_A \in B$
- $x \in B \implies -x \in B$
- $x, y \in B \implies x + y \in B$
- $x, y \in B \implies x \cdot y \in B$

Then $(B, +|_{B\times B}, \cdot\,|_{B\times B})$ is a ring.

Definition 3.3.4 (Multiplicative set)
A subset $S \subset A$ is said to be ***multiplicative*** *if*

- $1 \in S$
- $x, y \in S \implies xy \in S$

Further it is said to be ***saturated*** *if in addition*

$$x, y \in S \iff xy \in S$$

Definition 3.3.5 (Integral Domain)
A commutative ring A is said to be an ***integral domain*** *if it is not a zero-ring and it is* cancellative*, that is*

$$ab = ac, a \neq 0 \implies b = c.$$

Definition 3.3.6 (Reduced)
A commutative ring A is said to be ***reduced*** *if for all $a \in A$*

$$a^n = 0 \implies a = 0$$

Definition 3.3.7 (Unit / Group of Units)
An element $0 \neq a$ of a ring A is called a ***unit*** *if it has a two-sided multiplicative inverse.*

For A not a zero-ring, the set of units $A^\star$ forms a group under multiplication, called the ***group of units****.*

Definition 3.3.8 (Field)
A ***field*** *K is a commutative non-zero ring such that every non-zero element is a unit, so that $K^\star$ is a group under multiplication and $K^\star = K \setminus \{0\}$.*

Proposition 3.3.9
Every subring of a field K is an integral domain.

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Proof. Suppose $A \subset K$ is a subring. Suppose that $a, b \in A$ such that $ab = 0$. Suppose $a \neq 0$ then $a^{-1}ab = 0 \implies b = 0$. ∎

Note we have the implications

Corollary 3.3.10
Let A be a ring then we have the following implications

$$\textbf{field} \implies \textbf{integral domain} \implies \textbf{reduced}$$

Definition 3.3.11 (Ring homomorphism)
*A **ring homomorphism** $\varphi : A \to B$ is a mapping which is both a multiplicative (monoid) and additive (group) homomorphism*

- $\varphi(0_A) = 0_B$
- $\varphi(1_B) = 1_B$
- $\varphi(x + y) = \varphi(x) + \varphi(y)$
- $\varphi(xy) = \varphi(x)\varphi(y)$

*The **kernel** of φ is defined to be*

$$\ker(\varphi) = \{a \mid \varphi(a) = 0_B\}$$

Definition 3.3.12 (Ideal)
*A (two-sided) **ideal** $\mathfrak{a}$ of a ring A is a subset of A which is an additive subgroup and closed under multiplication by A :*

- $0_A \in \mathfrak{a}$
- $x, y \in \mathfrak{a} \implies x + y \in \mathfrak{a}$
- $x \in \mathfrak{a} \implies -x \in \mathfrak{a}$
- $x \in \mathfrak{a}, a \in A \implies ax, xa \in \mathfrak{a}$

*$\mathfrak{a}$ is said to be **proper** if $\mathfrak{a} \neq A$.*

Lemma 3.3.13 (Proper ideal)
An ideal $\mathfrak{a}$ is proper if and only if $1 \notin \mathfrak{a}$ if and only if $\mathfrak{a} \cap A^\star = \emptyset$.

Alternatively $\mathfrak{a} = A$ if and only if $1 \in \mathfrak{a}$ if and only if $\mathfrak{a} \cap A^\star \neq \emptyset$.

Proposition 3.3.14
Let $\varphi : A \to B$ be a ring homomorphism, then

a) The kernel $\ker(\varphi)$ is a two-sided ideal of A
b) The image $\varphi(A)$ is a subring of B
c) φ is injective if and only if $\ker(\varphi) = \{0\}$

Proposition 3.3.15 (Krull's Theorem)
Let A be a ring and $\mathfrak{a}$ a proper ideal. Then it is contained in a proper maximal ideal $\mathfrak{m}$.

In particular any non-unit $a \notin A^\star$ is contained in a maximal ideal.

Proposition 3.3.16 (Criteria to be a Field)
Let A be a ring. Then the following are equivalent

a) A is a field
b) A is not the zero-ring and the only proper ideal is $\{0\}$.

Proof. a) $\implies$ b). By definition A is not the zero-ring. Let $\mathfrak{a}$ be a proper ideal. By (3.3.13) $\mathfrak{a} \cap A^\star = \emptyset \implies \mathfrak{a} = \{0\}$ as required.

b) $\implies$ a). Suppose $0 \neq a \in A$ then the ideal $Aa = (a)$ is either $\{0\}$ or A. However $a = 1 \cdot a \in (a)$ which implies $(a) \neq \{0\}$ and therefore $(a) = A$. In particular there exists $a^{-1} \in A$ such that $a^{-1}a = 1_A$ and a is invertible. ∎

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3.3.2 Sums in a Ring

Proposition 3.3.17 (Distributivity for Sums in a Ring)
Let A be a ring, $r \in A$ and $(a_i)_{i\in I}$ a family in A with all but finitely many terms zero. Then $(ra_i)_{i\in I}$ and $(a_i r)_{i\in I}$ again have all but finitely many terms zero, and

$$r\sum_{i\in I} a_i = \sum_{i\in I} ra_i \qquad \left(\sum_{i\in I} a_i\right) r = \sum_{i\in I} a_i r$$

If $(b_j)_{j\in J}$ is another such family then $(a_i b_j)_{(i,j)\in I\times J}$ has all but finitely many terms zero, and

$$\left(\sum_{i\in I} a_i\right)\left(\sum_{j\in J} b_j\right) = \sum_{(i,j)\in I\times J} a_i b_j = \sum_{i\in I}\sum_{j\in J} a_i b_j$$

and likewise with the order of summation exchanged.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

By the distributive property $x \mapsto rx$ and $x \mapsto xr$ are homomorphisms of the additive group of A, so the first claims are (3.2.49) for these homomorphisms. If $a_i b_j \neq 0$ then $a_i \neq 0$ and $b_j \neq 0$, so $a_i b_j$ vanishes outside a product of two finite sets. Write $b := \sum_{j\in J} b_j$. Then by the first part applied to b and then to each a_i

$$\left(\sum_{i\in I} a_i\right) b = \sum_{i\in I} a_i b = \sum_{i\in I} a_i \sum_{j\in J} b_j = \sum_{i\in I}\sum_{j\in J} a_i b_j = \sum_{(i,j)\in I\times J} a_i b_j$$

where the last equality is (3.2.49). The other order of summation is identical. ∎

3.3.3 Modules I

Definition 3.3.18 (Module)
*Let A be a ring. A **left A-module** $(M, +, \cdot)$ is an abelian group $(M, +)$ together with a "multiplication" operation*

$$\cdot : A \times M \to M$$

which satisfies the following properties

- $(a \times_A a') \cdot x = a \cdot (a' \cdot x)$
- $(a +_A a') \cdot x = a \cdot x + a' \cdot x$
- $a \cdot (x + y) = a \cdot x + a \cdot y$

*Similarly a **right A-module** $(M, +, \cdot)$ is an abelian group $(M, +)$ together with a multiplication operation*

$$\cdot : M \times A \to M$$

- $(a \times_A a') \cdot x = (x \cdot a') \times_A a$
- $(a +_A a') \cdot x = a \cdot x + a' \cdot x$
- $a \cdot (x + y) = a \cdot x + a \cdot y$

*Considering the first property M is a left A-module iff it is a right A^{op} module in the obvious way. Therefore in the usual case that A is commutative the concepts coincide and we may speak simply of an **A-module**, though we almost always write the action on the left.*

Similarly almost all results for left A-modules carry over unchanged for right A-modules. In this case we may simply by refer to A-modules rather than state the result for both cases separately.

Definition 3.3.19 (Submodule)
*Let $(M, +, \cdot)$ be a left A-module. Then a subset $N \subset M$ is called an A-**submodule** if*

- *N is a subgroup of $(M, +)$*
- *$m \in N, a \in A \implies am \in N$*

Then $(N, +|_{N\times N}, \cdot|_{A\times N})$ is a left A-module. Similar definition applies for a right A-module.

Definition 3.3.20 (Module homomorphism)
Let $(M, +, \cdot), (N, +, \cdot)$ be left A-modules. A function $f : M \to N$ is an A-module homomorphism if

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- *It is an (additive) group homomorphism* $(M,+) \to (N,+)$.
- *It is A-linear;* $\forall a \in A, m \in M \quad f(a \cdot m) = a \cdot f(m)$

It may be verified that f *is bijective if and only if it's an isomorphism. In this way we have the following categories*

- A-**Mod** *the category of modules over a* commutative *ring* A
- ${}_A$**Mod** *the category of left* A*-modules*
- **Mod**$_A$ *the category if right* A*-modules*

Definition 3.3.21 (Kernel and Image)
The ***kernel*** *of a module homomorphism* f *is given by*

$$\ker(f) := \{m \in M \mid f(m) = 0\}$$

and the ***image*** *is given by*

$$\mathrm{Im}(f) = f(M)$$

Example 3.3.22 (Trivial Examples)
A ring A *is a left* A*-module over itself, denoted* A_s.

Definition 3.3.23 (Restriction of Scalars)
Let $\varphi : A \to B$ *a ring homomorphism and* M *a* B*-module. Then we may consider* M *as an* A*-module in the obvious way. Denote this by* $[M]_\varphi$.

Proposition 3.3.24 (Submodules constitute a lattice)
Let M *be an* A*-module then the collection* $\mathrm{SubMod}(M)$ *of* A*-submodules form a complete sub-lattice of* $\mathcal{P}(M)$ *with meet and join given by*

$$\bigwedge_{i\in I} N_i = \bigcap_{i\in I} N_i$$

and (the ***internal sum****)*

$$\bigvee_{i\in I} N_i = \bigcap_{N_i \subseteq N \leq M} N =: \sum_{i\in I} N_i = \left\{ \sum_{j\in J} n_j \;\middle|\; n_j \in N_j \quad \#J < \infty \right\}$$

Moreover it is the image of the closure operator $\langle - \rangle : \mathcal{P}(M) \to \mathcal{P}(M)$ *given by*

$$\langle X \rangle = \bigcap_{X \subseteq N} N = \left\{ \sum_j a_j x_j \;\middle|\; x_j \in X \right\}$$

Proof. The A-submodules of M naturally form a Moore family of subsets of M. By (2.1.41) they form a complete sub-lattice with the given form of meet and join. Furthermore it is the image of the given closure operator. The only non-trivial statement is the explicit form of $\sum_{i\in I} N_i$ TODO. ∎

Lemma 3.3.25
Let M *be a module. Then*

a) $\langle \bigcup_{i\in I} X_i \rangle = \sum_{i\in I} \langle X_i \rangle$
b) $\langle \bigcup_{i\in I} N_i \rangle = \sum_{i\in I} N_i$
c) $N_1 \subseteq N_2 \implies N_1 + N_2 = N_2$

Proof.
a) This follows from (2.1.44) applied to the closure operator $\langle - \rangle$
b) This follows from a) because $N_i = \langle N_i \rangle$
c) This follows from b) because $N_1 \cup N_2 = N_2$

∎

Definition 3.3.26 (Hom Sets)
A ***module homomorphism*** $\varphi : M \to N$ *is an additive group homomorphism which commutes with the* A *action*

$$\varphi(am) = a\varphi(m) \quad \forall a \in A \, m \in M$$

Denote the abelian group of A*-module homomorphisms*

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$$\mathrm{Hom}_A(M,N)$$

and the endomorphism ring

$$\mathrm{End}_A(M) \coloneqq \mathrm{Hom}_A(M,M)$$

When A is commutative then these have natural A-module and A-algebra structures respectively.

3.3.4 Operations on Ideals

For this section we assume A is a commutative ring.

Definition 3.3.27 (Product of ideal and module)
Let M be an A-module and $\mathfrak{a} \triangleleft A$ an ideal. Define

$$\mathfrak{a}M = \langle \mathfrak{a} \cdot M \rangle = \left\{ \sum_{i=1}^{n} a_i m_i \;\middle|\; a_i \in \mathfrak{a} \quad m_i \in M \right\}$$

Proposition 3.3.28 (Lattice of Ideals)
Let A be a ring and $\mathcal{I}(A)$ the set of ideals. Then $\mathcal{I}(A)$ forms a complete lattice ordered by inclusion with join and meets given by

$$\bigwedge_{i\in I} \mathfrak{a}_i = \bigcap_{i\in I} \mathfrak{a}_i$$

and

$$\bigvee_{i\in I} \mathfrak{a}_i = \bigcap_{\mathfrak{a}_i \subseteq \mathfrak{a}} \mathfrak{a} =: \sum_i \mathfrak{a}_i \coloneqq \left\{ \sum_i a_i \;\middle|\; a_i \in \mathfrak{a}_i \right\}$$

This induces a corresponding closure operator

$$\langle - \rangle : \mathcal{P}(A) \to \mathcal{I}(A)$$

given by

$$\langle X \rangle \coloneqq \bigcap_{X \subseteq \mathfrak{a}} \mathfrak{a} = \left\{ \sum_j a_j x_j \;\middle|\; a_j \in A \quad x_j \in X \right\}$$

Proposition 3.3.29
Let A be a ring and $\mathfrak{a}_i$ a family of ideals. Then

$$\langle \bigcup_{i\in I} \mathfrak{a}_i \rangle = \sum_{i\in I} \mathfrak{a}_i$$

Definition 3.3.30 (Product of ideals)
The product of two ideals $\mathfrak{ab}$ is

$$\mathfrak{ab} = \left\{ \sum_{i=1}^{n} a_i b_i \;\middle|\; a_i \in \mathfrak{a} \quad b_i \in \mathfrak{b} \right\}$$

and is itself an ideal.

Definition 3.3.31 (Coprime)
*We say two elements x, y of a commutative ring A are **co-prime** if $(x,y) = (1) \iff ax + by = 1$ for some $a, b \in A$*

We say a family of ideals $\{\mathfrak{a}_i\}_{i\in I}$ are co-prime if $\sum_{i\in I} \mathfrak{a}_i = A$.

Definition 3.3.32 (Principal Ideal)
*A **principal ideal** is an ideal generated by a single element*

$$(a) \coloneqq \langle \{a\} \rangle = Aa$$

Lemma 3.3.33
A principal ideal (a) is proper if and only if $a \notin A^\star$

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Definition 3.3.34 (Maximal Ideal)
*An ideal $\mathfrak{m} \triangleleft A$ is **maximal** if it is both proper and not contained in another proper ideal.*

Definition 3.3.35 (Prime Ideal)
*An ideal $\mathfrak{p} \triangleleft A$ is **prime** if it is both proper and satisfies the following property*

$$xy \in \mathfrak{p} \implies x \in \mathfrak{p} \vee y \in \mathfrak{p}$$

Definition 3.3.36 (Radical Ideal)
*An ideal $\mathfrak{a} \triangleleft A$ is **radical** if it satisfies the following property*

$$x^n \in \mathfrak{a} \implies x \in \mathfrak{a}$$

Proposition 3.3.37 (Maximal ideals exist)
Let A be a ring and $\mathfrak{a} \triangleleft A$ a proper ideal. Then it is contained in some maximal ideal $\mathfrak{m}$.

In particular there always exists a maximal ideal by considering $\mathfrak{a} = (0)$.

Proof. Simple application of Zorn's Lemma. ∎

Proposition 3.3.38 (Properties of prime ideals)
Let $\mathfrak{p}$ be a prime ideal and $\mathfrak{a}$, $\mathfrak{b}$ be ideals then the following are equivalent

a) $\mathfrak{a} \subseteq \mathfrak{p}$ or $\mathfrak{b} \subseteq \mathfrak{p}$
b) $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{p}$
c) $\mathfrak{ab} \subseteq \mathfrak{p}$

Proof. a) $\implies$ b) Follows because $\mathfrak{a} \cap \mathfrak{b} \subseteq \mathfrak{a}$

b) $\implies$ c) Follows because $\mathfrak{ab} \subseteq \mathfrak{a} \cap \mathfrak{b}$

c) $\implies$ a) If $\mathfrak{a} \nsubseteq \mathfrak{p}$, then choose $a \in \mathfrak{a} \setminus \mathfrak{p}$. By hypothesis $a\mathfrak{b} \subseteq \mathfrak{p}$ and since $\mathfrak{p}$ is prime $\mathfrak{b} \subseteq \mathfrak{p}$. ∎

Corollary 3.3.39 (Ideal version of primality)
Let $\mathfrak{p}$ be a proper ideal. Then $\mathfrak{p}$ is prime if and only if the following condition holds for all ideals $\mathfrak{a}$, $\mathfrak{b}$

$$\mathfrak{ab} \subseteq \mathfrak{p} \implies \mathfrak{a} \subseteq \mathfrak{p} \text{ or } \mathfrak{b} \subseteq \mathfrak{p}$$

In particular for all $k > 0$ we have

$$\mathfrak{a} \subseteq \mathfrak{p} \iff \mathfrak{a}^k \subseteq \mathfrak{p}$$

Proof. One direction has been shown in (3.3.38). Conversely suppose $fg \in \mathfrak{p}$ then apply the condition to the ideals (f) and (g) we find $f \in \mathfrak{p}$ or $g \in \mathfrak{p}$. ∎

Lemma 3.3.40 (Prime ideals are meet-prime)
Let $\mathfrak{p}$ be a prime ideal. Then

$$\bigcap_{i=1}^{n} \mathfrak{a}_i \subseteq \mathfrak{p} \implies \mathfrak{a}_i \subseteq \mathfrak{p} \text{ some } i = 1...n$$

in other words $\mathfrak{p}$ is meet-prime in the lattice of ideals.

Proof. Suppose $\mathfrak{a}_i \nsubseteq \mathfrak{p}$ for all i then there exists $x_i \in \mathfrak{a}_i \setminus \mathfrak{p}$. Then $x_1...x_n \in \bigcap_{i=1}^{n} \mathfrak{a}_i \subseteq \mathfrak{p}$ by hypothesis, so by primality $x_i \in \mathfrak{p}$ for some i, a contradiction. ∎

Lemma 3.3.41 (Generate prime ideals)
Let A be a ring, S a multiplicative set and $\mathfrak{b} \triangleleft A$ such that $\mathfrak{b} \cap S = \emptyset$ then

$$\mathcal{I} = \{\mathfrak{a} \mid \mathfrak{b} \subseteq \mathfrak{a} \quad \mathfrak{a} \cap S = \emptyset\}$$

has a maximal element, which is prime.

Proof. Since $\mathfrak{b} \in \mathcal{I}$ it is non-empty. By Zorn's Lemma it has a maximal element, $\mathfrak{p}$. We claim it is prime, for suppose $xy \in \mathfrak{p}$ and $x, y \notin \mathfrak{p}$. Then by maximality $\mathfrak{p} + (x)$ and $\mathfrak{p} + (y)$ intersect S. Therefore S intersects $(\mathfrak{p} + (x))(\mathfrak{p} + (y)) \subseteq \mathfrak{p}$, a contradiction. ∎

Definition 3.3.42 (Minimal prime)
*Let A be a ring and $\mathfrak{a} \triangleleft A$ a proper ideal. A prime ideal $\mathfrak{p}$ is a **minimal prime over** $\mathfrak{a}$ if it contains $\mathfrak{a}$, and every other such prime ideal contains $\mathfrak{p}$.*

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We say it is simply a ***minimal prime*** *if it is minimal over* (0).

Proposition 3.3.43 (Prime ideals are chain complete)
Let $\{\mathfrak{p}_i\}_{i\in I}$ *be a* ***chain*** *of prime ideals, then* $\bigcap_i \mathfrak{p}_i$ *and* $\bigcup_i \mathfrak{p}_i$ *are prime ideals.*

Proof. By (3.3.37) $\bigcup_i \mathfrak{p}_i$ is an ideal, and it's easily verified to be prime. Clearly $\bigcap_i \mathfrak{p}_i$ is an ideal. Suppose $a, b \notin \bigcap_i \mathfrak{p}_i$ then $a \notin \mathfrak{p}_j$ and $b \notin \mathfrak{p}_k$ with $j \leq k$. Then $b \notin \mathfrak{p}_j$, and $ab \notin \mathfrak{p}_j$ by primality, whence $ab \notin \bigcap_i \mathfrak{p}_i$. ■

Corollary 3.3.44 (Minimal primes exist)
Let A *be a ring and* $\mathfrak{a} \lhd A$ *be a proper ideal contained in a prime ideal* $\mathfrak{p}$*. Then there exists a minimal prime over* $\mathfrak{a}$ *contained in* $\mathfrak{p}$*.*

In particular there always exists a minimal prime over $\mathfrak{a}$ *and every prime ideal contains a minimal prime ideal.*

Proof. We may use (3.3.43) together with Zorn's Lemma. ■

Proposition 3.3.45
Let A *be a ring. Then the set* $\mathrm{Rad}(A)$ *of radical ideals forms a complete sub-lattice of the lattice of ideals* $\mathcal{I}(A)$*. This induces a closure operator*

$$\sqrt{-} : \mathcal{I}(A) \to \mathrm{Rad}(A)$$

given by

$$\sqrt{\mathfrak{a}} \coloneqq \bigcap_{\mathfrak{a}\subseteq\mathfrak{r}} \mathfrak{r} = \{x \mid x^n \in \mathfrak{a} \quad n > 0\}$$

The "join" is given by

$$\bigvee_{i\in I} \mathfrak{a}_i = \sqrt{\sum_i \mathfrak{a}_i}$$

In particular

a) $\mathfrak{a} \subseteq \sqrt{\mathfrak{a}}$
b) $\mathfrak{a} \subseteq \mathfrak{b} \implies \sqrt{\mathfrak{a}} \subseteq \sqrt{\mathfrak{b}}$
c) $\sqrt{\sqrt{\mathfrak{a}}} = \sqrt{\mathfrak{a}}$

Proof. The set of radical ideals is closed under arbitrary intersections (which are meets in the lattice $\mathcal{I}(A)$). Therefore by (2.1.41) it forms a complete sub-lattice with meet given by intersection of ideals.

It also shows that $\sqrt{-}$ as defined is a closure operator with image $\mathrm{Rad}(A)$, which demonstrates the required properties.

Finally we just need to show that $I' \coloneqq \{x \mid x^n \in \mathfrak{a} \quad n > 0\}$ is equal to $\sqrt{\mathfrak{a}}$. Firstly it's an ideal for if $x, y \in I'$ then $x^n \in \mathfrak{a}$ and $y^m \in \mathfrak{a}$, so we may show that $(x+y)^{n+m} \in \mathfrak{a}$ whence $x + y \in I'$. Similarly $a \in A$ and $x \in I'$ implies $(ax)^n = a^n x^n \in I'$. It's radical for suppose $x^m \in I'$ then $x^{mn} = (x^m)^n \in \mathfrak{a}$ by definition whence $x \in I'$. As it contains $\mathfrak{a}$ we find that $\sqrt{\mathfrak{a}} \subseteq I'$. Let $\mathfrak{r}$ be another radical ideal containing $\mathfrak{a}$ then $x \in I' \implies x^n \in \mathfrak{a} \implies x^n \in \mathfrak{r} \implies x \in \mathfrak{r}$. Therefore the reverse inclusion follows. ■

Proposition 3.3.46 (Prime Nullstellensatz)
The radical of an ideal satisfies

$$\sqrt{\mathfrak{a}} = \bigcap_{\mathfrak{a}\subseteq\mathfrak{p}:\mathfrak{p}\text{ prime}} \mathfrak{p}$$

Further the intersection may be taken over all minimal *primes over* $\mathfrak{a}$*.*

Proof. Suppose $x \in \sqrt{\mathfrak{a}}$ and $\mathfrak{p} \supseteq \mathfrak{a}$. Then $x^n \in \mathfrak{p} \implies x \in \mathfrak{p}$. Therefore $\sqrt{\mathfrak{a}} \subseteq \bigcap_{\mathfrak{a}\subseteq\mathfrak{p}} \mathfrak{p}$. Conversely suppose $x \notin \sqrt{\mathfrak{a}}$ then $S \coloneqq \{1, x, x^2, \ldots\}$ is a proper multiplicative set such that $S \cap \mathfrak{a} = \emptyset$. By (3.3.41) there is a prime ideal $\mathfrak{p}$ containing $\mathfrak{a}$ which does not intersect S. Therefore $x \notin$ RHS as required. ■

Proposition 3.3.47 (Properties of Radical Ideals)
Let $\mathfrak{a}$, $\mathfrak{b}$ *be ideals then*

a) $\sqrt{\mathfrak{a}^k} = \sqrt{\mathfrak{a}}$ *for* $k > 0$
b) $\sqrt{\sum_i \mathfrak{a}_i} = \sqrt{\sum_i \sqrt{\mathfrak{a}_i}}$
c) $\sqrt{\mathfrak{a}} = A \iff \mathfrak{a} = A$
d) $\sum_i \mathfrak{a}_i = A \iff \sum_i \sqrt{\mathfrak{a}_i} = A$
e) $\sum_{i=1}^n \mathfrak{a}_i^{k_i} = A \iff \sum_{i=1}^n \mathfrak{a}_i = A \quad k_i > 0.$

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Proof.

a) This may be shown by direct calculation or combining (3.3.46) and (3.3.38).
b) This follows by applying (2.1.44) to the closure operator $\sqrt{-}$.
c) $\sqrt{\mathfrak{a}} = A \iff 1 \in \sqrt{\mathfrak{a}} \iff 1 \in \mathfrak{a} \iff \mathfrak{a} = A$
d) This follows from combining c) and b)
e) This follows from d), b) and a)

■

Lemma 3.3.48 (Ideal Finitely Generated by Nilpotents is Nilpotent)
Let A be a ring with ideals $\mathfrak{a}, \mathfrak{b}$ such that $\sqrt{\mathfrak{a}} \subseteq \mathfrak{b}$ and $\mathfrak{b}$ finitely generated. Then there exists an integer $n > 0$ such that $\mathfrak{b}^n \subseteq \mathfrak{a}$.

Definition 3.3.49 (Extended and contracted ideals)
*Let $\varphi : A \to B$ be a homomorphism and $\mathfrak{a}$ (resp. $\mathfrak{b}$) be an ideal of A (resp. B). Define the **contraction** (resp. **extension**) ideals as follows*

$$\mathfrak{b}^c := \varphi^{-1}(\mathfrak{b})$$
$$\mathfrak{a}^e := \varphi(\mathfrak{a})B := \langle \varphi(\mathfrak{a}) \rangle = \left\{ \sum_i b_i \varphi(a_i) \;\middle|\; a_i \in \mathfrak{a} \right\}$$

*An ideal is said to be **contracted** (resp. **extended**) if it is of the form $\mathfrak{b}^c$ (resp. $\mathfrak{a}^e$)*

Proposition 3.3.50 (Operations on ideals)
Let $\varphi : A \to B$ a ring homomorphism and $\mathfrak{a} \lhd A$, $\mathfrak{b} \lhd B$ ideals then

a) $\mathfrak{b}^c \lhd A$ and $\mathfrak{a}^e \lhd B$
b) $\mathfrak{b}^c$ proper if and only if $\mathfrak{b}$ is proper
c) $\mathfrak{b}^{ce} \subseteq \mathfrak{b}$ and $\mathfrak{a} \subseteq \mathfrak{a}^{ec}$
d) $\mathfrak{a}^{ece} = \mathfrak{a}^e$ and $\mathfrak{b}^{cec} = \mathfrak{b}^c$
e) $\mathfrak{b}^{ce} = \mathfrak{b} \iff \mathfrak{b}$ is an extended ideal $\iff \mathfrak{b} \subseteq \mathfrak{b}^{ce}$
f) $\mathfrak{a}^{ec} = \mathfrak{a} \iff \mathfrak{a}$ is a contracted ideal $\iff \mathfrak{a}^{ec} \subseteq \mathfrak{a}$
g) $\sqrt{\mathfrak{b}^c} = \left(\sqrt{\mathfrak{b}}\right)^c$
h) $\left(\sqrt{\mathfrak{b}^c}\right)^e \subseteq \sqrt{\mathfrak{b}}$ with equality when φ is surjective

When φ is surjective every ideal $\mathfrak{b} \lhd B$ is extended, and the contracted ideals are precisely the ideals containing $\ker(\varphi)$.

Proof. We prove each in turn

a)-c) Straightforward

d) By the previous step $\mathfrak{b}^{ce} \subseteq \mathfrak{b} \implies (\mathfrak{b}^{ce})^c \subseteq \mathfrak{b}^c$, similarly $\mathfrak{b}^c \subseteq (\mathfrak{b}^c)^{ec}$. The other relation is similar.

e)-f) These follow from c) and d)

g) $x \in \left(\sqrt{\mathfrak{b}}\right)^c \iff \varphi(x) \in \sqrt{\mathfrak{b}} \iff \varphi(x)^n \in \mathfrak{b} \iff \varphi(x^n) \in \mathfrak{b} \iff x^n \in \mathfrak{b}^c \iff x \in \sqrt{\mathfrak{b}^c}$

h) By c) and g) we find $\left(\sqrt{\mathfrak{b}^c}\right)^e = \left(\sqrt{\mathfrak{b}}\right)^{ce} \subseteq \sqrt{\mathfrak{b}}$. We will show that when φ is surjective every ideal is extended, in which case the equality follows from e).

Suppose that φ is surjective. Then by e) we only need to show that $\mathfrak{b} \subseteq \mathfrak{b}^{ce}$ for every ideal $\mathfrak{b}$. Let $y \in \mathfrak{b}$ then $y = \varphi(x)$, whence $x \in \mathfrak{b}^c$ and $y \in \mathfrak{b}^{ce}$. ■

Proposition 3.3.51
Let $\varphi : A \to B$ be a ring homomorphism and $\mathfrak{a}_1, ..., \mathfrak{a}_n$ ideals of A. Then

$$\left(\sum_{i=1}^n \mathfrak{a}_i\right)^e = \sum_{i=1}^n \mathfrak{a}_i^e$$

Similarly

$$\left(\prod_{i=1}^n \mathfrak{a}_i\right)^e = \prod_{i=1}^n \mathfrak{a}_i^e$$

Proof. Evidently $\varphi(\mathfrak{a}_i) \subseteq \varphi\left(\sum_{i=1}^n \mathfrak{a}_i\right) \subseteq \left(\sum_{i=1}^n \mathfrak{a}_i\right)^e$ and therefore $\mathfrak{a}_i^e \subseteq \left(\sum_{i=1}^n \mathfrak{a}_i\right)^e \implies \sum_{i=1}^n \mathfrak{a}_i^e \subseteq \left(\sum_{i=1}^n \mathfrak{a}_i\right)^e$.

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Similarly $\varphi\left(\sum_{i=1}^n \mathfrak{a}_i\right) \subseteq \sum_{i=1}^n \varphi(\mathfrak{a}_i) \subseteq \sum_{i=1}^n \mathfrak{a}_i^e$ whence the reverse inclusion follows.

The second statement follows similarly. ■

Corollary 3.3.52
Let $\varphi : A \to B$ be a ring homomorphism then extension and contraction constitute a monotone Galois connection

$$\{\mathfrak{a} \lhd A\} \longleftrightarrow \{\mathfrak{b} \lhd B\}$$

and therefore is order-preserving and satisfies the adjoint property

$$\mathfrak{a} \subseteq \mathfrak{b}^c \iff \mathfrak{a}^e \subseteq \mathfrak{b}$$

is satisfied.

Proof. Extension and contraction satisfy conditions of (2.1.50) by (3.3.50) c) and d) ■

Corollary 3.3.53
Let $\varphi : A \to B$ be a ring homomorphism then there is a order-preserving bijection between "contracted" and "extended ideals"

$$\{\mathfrak{a} \lhd A \mid \mathfrak{a} \text{ contracted}\} \longleftrightarrow \{\mathfrak{b} \lhd B \mid \mathfrak{b} \text{ extended}\}$$

which restricts to proper ideals.

Proof. We've shown that $\mathfrak{a}$ (resp. $\mathfrak{b}$) is contracted (resp. extended) if and only if the given maps are mutually inverse. Note that $\mathfrak{b}$ is proper implies $\mathfrak{b}^c$ is proper. Furthermore $\mathfrak{b}^c$ proper implies $\mathfrak{b}^{ce} \subseteq \mathfrak{b}$ is proper. Therefore it restricts to proper ideals. ■

Proposition 3.3.54 (Inverse image of maximal / prime ideals)
Let $\varphi : A \to B$ be a morphism then

- *$\mathfrak{q} \lhd B$ prime $\implies \varphi^{-1}(\mathfrak{q})$ prime*
- *$\mathfrak{n} \lhd B$ maximal and φ surjective $\implies \varphi^{-1}(\mathfrak{n})$ is maximal*

Proposition 3.3.55
Consider maps $\varphi : A \to B$ and $\psi : B \to C$ and an ideal $\mathfrak{a} \lhd A$. Then extension of ideals is transitive, that is

$$\psi(\varphi(\mathfrak{a})B)C = (\psi \circ \varphi)(\mathfrak{a})C$$

3.3.5 Quotient Rings

Proposition 3.3.56 (Quotient Ring)
Let $(A, +, \cdot)$ be a ring and $\mathfrak{a}$ an ideal. As $\mathfrak{a}$ is an additive subgroup we may consider the quotient group $(A/\mathfrak{a}, +)$. For an element $a \in A$ write $a + \mathfrak{a}$ for the coset $[a]_\mathfrak{a} \in A/\mathfrak{a}$. There is a well-defined multiplicative law of composition

$$\cdot : A/\mathfrak{a} \times A/\mathfrak{a} \to A/\mathfrak{a}$$

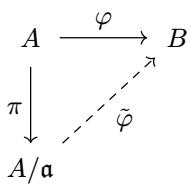
$$(a + \mathfrak{a}) \cdot (b + \mathfrak{a}) \to (a \cdot b + \mathfrak{a})$$

which makes $(A/\mathfrak{a}, +, \cdot)$ into a ring. Further there is a canonical surjective ring homomorphism

$$\pi : A \to A/\mathfrak{a}$$

with the following properties

- $\ker(\pi) = \mathfrak{a}$
- *Every morphism $\varphi : A \to B$ such that $\mathfrak{a} \subseteq \ker(\varphi)$, factors uniquely through π.*



- $\ker(\tilde{\varphi}) = \ker(\varphi)/\mathfrak{a}$
- *$\tilde{\varphi}$ is injective if and only if* $\ker(\varphi) = \mathfrak{a}$
- *$\tilde{\varphi}$ is surjective if and only if φ is surjective*

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For an ideal $\mathfrak{b} \supseteq \mathfrak{a}$ define the corresponding quotient ideal

$$\mathfrak{b}/\mathfrak{a} := \{b + \mathfrak{a} \mid b \in \mathfrak{b}\} = \pi(\mathfrak{b})$$

This induces a bijective, order-preserving correspondence of ideals

$$\{\mathfrak{b}' \lhd A/\mathfrak{a}\} \underset{\pi(-)}{\overset{\pi^{-1}(-)}{\rightleftarrows}} \{\mathfrak{b} \lhd A \mid \mathfrak{a} \subseteq \mathfrak{b}\}$$

under which maximal (resp. prime, radical) ideals of A containing $\mathfrak{a}$ correspond to maximal (resp. prime, radical) ideals of $A/\mathfrak{a}$.

Corollary 3.3.57 (Isomorphism Theorem)
Let $\varphi : A \to B$ be a ring homomorphism. Then this induces a canonical isomorphism

$$A/\ker(\varphi) \cong \varphi(A) \subset B$$

Corollary 3.3.58 (Second Isomorphism Theorem)
Let $\mathfrak{b}$,$\mathfrak{a}$ be ideals then there is a unique morphism making the diagram commute

$$\begin{array}{ccc} A & \xrightarrow{\pi} & A/\mathfrak{a} \\ {\scriptstyle\pi}\downarrow & \swarrow & \downarrow{\scriptstyle\pi} \\ A/(\mathfrak{a}+\mathfrak{b}) & \xrightarrow{\sim} & (A/\mathfrak{a})/((\mathfrak{a}+\mathfrak{b})/\mathfrak{a}) \end{array}$$

which is in fact an isomorphism. If $\mathfrak{a} + \mathfrak{b} \subseteq \mathfrak{c}$ this restricts to an isomorphism of $A/\mathfrak{b}$-modules

$$\begin{array}{ccc} \mathfrak{c} & \xrightarrow{\pi} & \mathfrak{c}/\mathfrak{a} \\ {\scriptstyle\pi}\downarrow & \swarrow & \downarrow{\scriptstyle\pi} \\ \mathfrak{c}/(\mathfrak{a}+\mathfrak{b}) & \xrightarrow{\sim} & (\mathfrak{c}/\mathfrak{a})/((\mathfrak{a}+\mathfrak{b})/\mathfrak{a}) \end{array}$$

Proposition 3.3.59 (Criteria for Maximal, Prime and Reduced)
Let $\mathfrak{a} \lhd A$ then $\mathfrak{a}$ is

- ***maximal** if and only if $A/\mathfrak{a}$ is a field*
- ***prime** if and only if $A/\mathfrak{a}$ is an integral domain*
- ***radical** if and only if $A/\mathfrak{a}$ is reduced*

Proof. Suppose $\mathfrak{a}$ is maximal then it is by definition proper so $A/\mathfrak{a}$ is not the zero-ring. By (3.3.56) then $A/\mathfrak{a}$ has no proper non-zero ideals and so by (3.3.16) is a field.

Conversely if $A/\mathfrak{a}$ is a field it is by definition not the zero ring, and so by (3.3.56) $\mathfrak{a}$ is proper. Furthermore by the same result $\mathfrak{a}$ is maximal.

Suppose $\mathfrak{a}$ is prime and $\overline{x} \cdot \overline{y} = 0$. Then by definition $\overline{x \cdot y} = 0 \implies x \cdot y \in \mathfrak{a} \implies x \in \mathfrak{a}$ or $y \in \mathfrak{a} \implies \overline{x} = 0$ or $\overline{y} = 0$. This shows that $A/\mathfrak{a}$ is an integral domain. The converse is similar. ∎

Corollary 3.3.60
Let $\mathfrak{a} \lhd A$ be a proper ideal, then the following implications hold

$$\textbf{maximal} \implies \textbf{prime} \implies \textbf{radical}$$

Proof. This follows by combining (3.3.10) and (3.3.59). ∎

Corollary 3.3.61 (Field Morphisms are injective)
Let $\varphi : k \to B$ be a homomorphism from a field to a non-zero ring. Then φ is injective.

Proof. $\ker(\varphi)$ is an ideal. As $\varphi(1_k) = 1_B$ and $0_B \neq 1_B$ then $\ker(\varphi) \neq k$. Since the only ideals are (0) and k we see $\ker(\varphi) = \{0\}$ and φ is injective. ∎

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3.3.6 Chinese Remainder Theorem

Proposition 3.3.62 (Chinese Remainder Theorem)
Let A be a commutative ring with ideals $\mathfrak{a}_1, ..., \mathfrak{a}_n$ which are pairwise coprime (i.e. $\mathfrak{a}_i + \mathfrak{a}_j = A$ for all $1 \leq i < j \leq n$). Then there is a canonical isomorphism

$$A/\bigcap_{i=1}^{n} \mathfrak{a}_i \longrightarrow A/\mathfrak{a}_1 \times ... \times A/\mathfrak{a}_n$$

Further we have the following identity

$$\bigcap_{i=1}^{n} \mathfrak{a}_i = \prod_{i=1}^{n} \mathfrak{a}_i$$

Proof. Consider the canonical ring homomorphism

$$\varphi : A \longrightarrow A/\mathfrak{a}_1 \times ... \times A/\mathfrak{a}_n.$$

Trivially it has kernel $\bigcap_{i=1}^{n} \mathfrak{a}_i$. Therefore by (3.3.56) it is sufficient to show it is surjective. First we show that $e_1 := (1, 0, ..., 0)$ is in the image of φ. For we have $x_i + y_i = 1$ for $x_i \in \mathfrak{a}_1$, $y_i \in \mathfrak{a}_i$ and $1 < i \leq n$. Take $f_1 = y_2 ... y_n \in \bigcap_{i=2}^{n} \mathfrak{a}_i$; since $y_i = 1 - x_i$, $f_1 = \prod_{i=2}^{n}(1 - x_i) \in 1 + \mathfrak{a}_1$ (as each $x_i \in \mathfrak{a}_1$), so combined with $f_1 \in \bigcap_{i=2}^{n} \mathfrak{a}_i$, $\varphi(f_1) = e_1$ as required.

Similarly there exists f_i such that $\varphi(f_i) = (0, ..., 1, ..., 0) =: e_i$. Finally for $\overline{x_1} \in A/\mathfrak{a}_1, ..., \overline{x_n} \in A/\mathfrak{a}_n$ then

$$\varphi(f_1 x_1 + ... + f_n x_n) = (\overline{x_1}, ..., \overline{x_n})$$

For the final statement we may assume by induction that

$$\bigcap_{i=1}^{n-1} \mathfrak{a}_i = \prod_{i=1}^{n-1} \mathfrak{a}_i$$

We have equations

$$x_i + y_i = 1 \quad x_i \in \mathfrak{a}_i, y_i \in \mathfrak{a}_n$$

Then

$$x_1 ... x_{n-1} = \prod_{i=1}^{n-1}(1 - y_i) \in 1 + \mathfrak{a}_n$$

so we may conclude

$$\bigcap_{i=1}^{n-1} \mathfrak{a}_i + \mathfrak{a}_n = A.$$

This then reduces to the case $n = 2$. Suppose $x \in \mathfrak{a} \cap \mathfrak{b}$, by assumption $1 = y + z$ for $y \in \mathfrak{a}$ and $z \in \mathfrak{b}$. Then $x = x(y + z) \in \mathfrak{ab}$, so $\mathfrak{a} \cap \mathfrak{b} \subset \mathfrak{ab}$. The reverse inclusion is obvious. ∎

Proposition 3.3.63 (Chinese Remainder Theorem II)
Let A be a commutative ring with $\mathfrak{a}_1, ..., \mathfrak{a}_n$ co-prime ideals. For any positive integers $k_1, ..., k_n$ there is a canonical isomorphism

$$A/\bigcap_{i=1}^{n} \mathfrak{a}_i^{k_i} \longrightarrow A/\mathfrak{a}_1^{k_1} \times ... \times A/\mathfrak{a}_n^{k_n}$$

and

$$\bigcap_{i=1}^{n} \mathfrak{a}_i^{k_i} = \prod_{i=1}^{n} \mathfrak{a}_i^{k_i}$$

Proof. By (3.3.47) the ideals $\mathfrak{a}_1^{k_1}, ..., \mathfrak{a}_n^{k_n}$ are pairwise co-prime. Therefore we may apply (3.3.62). ∎

3.3.7 Irreducible and Reduced rings

We say an element x is nilpotent if $x^n = 0$. By (3.3.45) these form an ideal.

Definition 3.3.64 (Nilradical)
*Define the **nilradical** to be the set (ideal) of nilpotents*

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$$N(A) := \sqrt{(0)} \overset{(3.3.46)}{=} \bigcap_{\mathfrak{p}} \mathfrak{p}$$

Clearly A is reduced if and only if $N(A) = \{0\}$.

We also make the following definition

Definition 3.3.65 (Irreducible)
*Let A be a ring. We say A is **irreducible** if $N(A)$ is prime.*

The notion of irreducible ring is related to the notion of minimal primes

Proposition 3.3.66
A ring A is irreducible if and only if it has a unique minimal prime ideal. In this case it is equal to $N(A)$.

Proof. First we note that every prime ideal contains $N(A)$. If A is irreducible then by definition $N(A)$ is prime and it is therefore the unique minimal prime ideal.

Conversely if $\mathfrak{p}_0$ is the unique minimal prime ideal then by (3.3.44) it is contained in every prime ideal. Therefore (3.3.64) $N(A) = \bigcap_{\mathfrak{p}} \mathfrak{p} = \mathfrak{p}_0$ is prime and A is irreducible. ∎

Proposition 3.3.67 (Integral Domain $\iff$ Reduced and Irreducible)
Let A be a ring. Then the following are equivalent

a) A is an integral domain
b) A is reduced and has a unique minimal prime
c) (0) is prime
d) A is reduced and irreducible.

The following may be useful

Proposition 3.3.68
Let A be a ring then the sum of an invertible and nilpotent element is again invertible

$$A^\star + N(A) \subseteq A^\star$$

Proof. For if

$$a = u + n = u(1 - (u^{-1})(-n))$$

with $n \in N(A)$ and $u \in A^\star$. Then way may reduce to the case $1 - n$ and observe that

$$(1-n)^{-1} = \sum_{i=0}^{\infty} n^i$$

which by assumption is a finite sum. ∎

3.3.8 Algebra over a Commutative Ring

For what follows let A be a commutative ring.

Definition 3.3.69 (Algebra (over a commutative ring))
An algebra over A (or an A-algebra) is a pair (i_B, B) where B is a (not necessarily commutative) ring and $i_B : A \to B$ is a ring homomorphism.

We call i_B the structural morphism and write $a \cdot b := i_B(a)b$

Morphisms of A-algebras are the ring homomorphisms $\varphi : B \to C$ such that $\varphi \circ i_B = i_C$. This then constitutes a category $\mathbf{Alg}_A$.

If k is a field an algebra over k is referred to as a k-algebra.

Definition 3.3.70 (Sub-algebra)
Let (i_B, B) be an A-algebra. A sub-algebra C is a subring C of B for which

$$a \in A \quad c \in C \implies i_B(a)c \in C$$

Example 3.3.71 (Algebra over commutatve sub-ring)
If $A \subset B$ is a commutative sub-ring, then B is naturally a A-algebra.

The polynomial ring $A[X]$ is naturally an A-algebra

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Definition 3.3.72 (Algebra generated by a set)
Let B be an A-algebra. The collection of A-subalgebras forms a Moore family. Therefore by (2.1.41) there is a canonical closure operator

$$A[-] : \mathcal{P}(B) \to \mathrm{SubAlg}_A(B)$$

which we denote by $A[S]$ for $S \subset B$. A more explicit characterization when S is finite is given in the Polynomial Rings in Many Variables subsection. More generally we have

$$A[S] = \bigcup_{S' \subset S | S' \text{ finite}} A[S']$$

Proposition 3.3.73
Let $B = A[S]$ be an A-algebra and $\mathfrak{a}$ a sub-A-module of B. Then $\mathfrak{a}$ is an ideal if and only if

$$s \in S \implies s\mathfrak{a} \subseteq \mathfrak{a}$$

Proof. One direction is obvious. Suppose the condition given holds, and define

$$B' := \{b \in B \mid b\mathfrak{a} \subseteq \mathfrak{a}\}$$

Then clearly $S \subseteq B'$. It's easy to show that B' is a sub-A-algebra of B, so $B' = B$ and $\mathfrak{a}$ is an ideal. ∎

3.3.9 Finite-type Algebras

Definition 3.3.74 (Finite algebra)
An A-algebra B is finite *if it is finite as an A-module.*

Definition 3.3.75 (Finitely generated algebra)
An A-algebra B is finitely generated (or of finite type) if there exists an integer $n \in \mathbb{N}$ and a surjection of A-algebras

$$A[X_1, ..., X_n] \to B$$

the images of X_i are the generators.

Proposition 3.3.76
Let C be a finitely-generated B algebra and B a finitely-generated A-algebra. Then C is finitely-generated as an A-algebra.

Proof. Suppose $C = B[c_1, ..., c_n]$. Then by definition $c_i = F_i(b_1, ..., b_m)$ for some $F_i \in A[X_1, ..., X_m]$. Then we may take as a generating set

$$b_1^{\lambda_1} ... b_m^{\lambda_m}$$

where $\lambda \in \mathbb{N}^m$ runs through all the exponents of non-zero coefficients of the F_i. ∎

3.3.10 Bimodules

For applications it is often useful to have multiple rings acting on a single module ("bimodule"). For example an A-algebra B naturally has an action from both A and B. It may also allow us to generalise results which would otherwise only hold in the commutative case.

Definition 3.3.77 (Bimodule)
*Let A, B be rings. We say an abelian group M is a (A, B)-**bimodule** if it is both a left A-module and a right B-module for which the two actions commute*

$$a \cdot (m \cdot b) = (a \cdot m) \cdot b \quad \forall a \in A, b \in B, m \in M$$

We may denote this by ${}_AM_B$ in order to emphasise the actions. If homomorphisms are defined in the obvious way then these constitute a category ${}_A\mathbf{Mod}_B$. Recall that every abelian group is automatically both a left and right $\mathbb{Z}$-module with the following action for $N \in \mathbb{Z}$

$$N \cdot m = m \cdot N := \begin{cases} \underbrace{m + ... + m}_{N \text{ times}} & N \geq 0 \\ -(m \cdot (-N)) & N < 0 \end{cases}$$

Therefore we have the following generalisations

- *A right B-module is precisely a $(\mathbb{Z}, B)$-bimodule*

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- *A left A-module is precisely a $(A,\mathbb{Z})$-bimodule*
- *When A is commutative an A-module has a well-defined (A,A)-bimodule structure by defining*

$$a\cdot m\cdot a' := a'a\cdot m = aa'\cdot m$$

  *in other words A-**Mod** is (equivalent to) a full subcategory of ${}_A\mathbf{Mod}_A$.*
- *A ring A has an obvious (A,A)-bimodule structure by associativity of the multiplication operation*

We will understand by the notation N_B that N is a $(\mathbb{Z},B)$-bimodule and by ${}\{\}_A M$ that M is a $(A,\mathbb{Z})$-bimodule.

We generalize slightly the consideration of the third bullet point

Proposition 3.3.78 ((A,A)-Bimodule $\equiv$ A-module)
Let $\varphi : A\to B$ be a homomorphism of commutative rings and Z a B-module. Then Z is naturally a (B,A)-module with the following action

$$b\cdot m\cdot a := \varphi(a)b\cdot m \qquad (\star)\quad \forall a\in A, b\in B, m\in Z$$

Suppose M is another (B,A)-bimodule satisfying $(\star)$ then we may identify the (B,A)-bimodule homomorphisms and the left B-module homomorphisms.

$$\mathrm{Hom}(M,\,{}_BZ_A)\xrightarrow{\sim}\mathrm{Hom}({}_BM,\,{}_BZ)$$

In particular when $\varphi = 1_A$ and $B=A$ we may identify (A,A)-bimodules and A-modules, as well as the corresponding homomorphisms. Explicitly for M,Z A-modules these have well-defined (A,A)-bimodule structures and there is a bijection

$$\mathrm{Hom}({}_AM_A,\,{}_AZ_A)\xrightarrow{\sim}\mathrm{Hom}(M,Z)$$

Proof. We may show that the right A-module structure on Z is well-defined because B is commutative and using the associativity of the B-module action. Similarly the associativity of the B-module action ensures that the actions commute and so form a (B,A)-bimodule structure.

Suppose $\theta : M\to Z$ is a (left) B-module homomorphism. Then

$$\theta(m\cdot a)=\theta(\varphi(a)\cdot m)=\varphi(a)\theta(m)=\theta(m)\cdot a$$

so it is (B,A)-bilinear as required. The converse is clear, so we see that the sets are equal as subsets of the set of abelian group homomorphisms.

The case $\varphi = 1_A$ is immediate. ∎

We generalize the notion of "Hom-Set"

Definition 3.3.79 (Hom Functors)
Let A,B,C be arbitrary rings. Then we have an enriched hom functor for ${}_A\mathbf{Mod}_B$

$$\mathrm{Hom} : {}_A\mathbf{Mod}_B^{op}\times{}_A\mathbf{Mod}_B\to\mathbf{AbGrp}$$

In order to generalise the Tensor-Hom adjunction we consider the following functors ("right-linear" and "left-linear" respectively)

$$\mathrm{RHom} : {}_B\mathbf{Mod}_A^{op}\times{}_C\mathbf{Mod}_A\to{}_C\mathbf{Mod}_B$$
$$\mathrm{LHom} : {}_A\mathbf{Mod}_B^{op}\times{}_A\mathbf{Mod}_C\to{}_B\mathbf{Mod}_C$$

For example if $\psi\in\mathrm{RHom}({}_BM_C,\,{}_AN_C)$ then the (A,B)-bimodule action is defined to be

$$(a\psi b)(m) := a\psi(bm)$$

and similarly if $\varphi\in\mathrm{LHom}({}_AM_B,\,{}_AN_C)$ then

$$(b\varphi c)(m) := \varphi(mb)c$$

which we may verify satisfies the axioms of a bimodule. In order to standardize notation we may write Hom_A in place of RHom *and* LHom.

If $B=C=\mathbb{Z}$ then we recover the simple case (3.3.26).

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*If A is a commutative ring then we've observed that A-**Mod** is a full subcategory of ${}_A\mathbf{Mod}_A$ and the functors* RHom, LHom *become equal to the simple case (3.3.26) when restricted to this subcategory.*

3.3.11 Module Direct Product and Sum

Definition 3.3.80 (External Direct Product / Sum)
*Let A be a ring and $\{M_i\}_{i\in I}$ a family of A-modules. Define the **external direct product** as the set of ordered tuples indexed over I*

$$\prod_{i\in I} M_i \coloneqq \left\{ (m_i)_{i\in I} \;\middle|\; m_i \in M_i \right\}$$

*with the obvious module operations. The **external direct sum** is the subset of tuples for which all but finitely many elements are zero. We denote this as follows*

$$\bigoplus_{i\in I} M_i$$

Clearly when I is finite then these are equal.

Proposition 3.3.81 (Categorical Product)
Let $\{M_i\}_{i\in I}$ be a family of left A-modules and consider the family of projections

$$\pi_i : \prod_{i\in I} M_i \to M_i$$

For Z an (A,C)-bimodule there is a natural isomorphism of left C-modules

$$\begin{aligned} \mathrm{Hom}_A\left(Z, \prod_{i\in I} M_i\right) &\xrightarrow{\sim} \prod_{i\in I} \mathrm{Hom}_A(Z, M_i) \\ \theta &\to (\pi_i \circ \theta)_{i\in I} \end{aligned}$$

This in particular includes the case $A = C$ is commutative.

Proposition 3.3.82 (Categorical Coproduct)
Let $\{M_i\}_{i\in I}$ be a family of left A-modules and consider the family of inclusions

$$u_i : M_i \to \bigoplus_{i\in I} M_i$$

For Z a (A,C)-bimodule there is a natural isomorphism of right C-modules

$$\begin{aligned} \mathrm{Hom}_A\left(\bigoplus_{i\in I} M_i, Z\right) &\xrightarrow{\sim} \prod_{i\in I} \mathrm{Hom}_A(M_i, Z) \\ \theta &\to (\theta \circ u_i)_{i\in I} \end{aligned}$$

This in particular includes the case $A = C$ is commutative.

Corollary 3.3.83
Let $(M_i)_{i\in I}$ and $(N_j)_{j\in J}$ be families of left A-modules. Then there is an isomorphism of abelian groups

$$\begin{aligned} \mathrm{Hom}_A\left(\bigoplus_{i\in I} M_i, \prod_{j\in J} N_j\right) &\cong \prod_{(i,j)\in I\times J} \mathrm{Hom}_A(M_i, N_j) \\ \psi &\to (\pi_j \circ \psi \circ u_i)_{(i,j)} \end{aligned}$$

where $\pi_j : \prod_{j\in J} N_j \to N_j$ and $u_i : M_i \to \bigoplus_{i\in I} M_i$ are the canonical projections and injections respectively.

These maps are A-linear if A is commutative.

Definition 3.3.84 (Free Module)
An A-module M is

- ***free** if it is isomorphic to*

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$$\bigoplus_{i\in A} A =: A^{(I)}$$

for some indexing set I

- ***finite free*** *if it is free with respect to a* finite *indexing set I*

Under the isomorphism $M \to \bigoplus_{i\in I} A$ the set of elements $\{m_i\}_{i\in I}$ corresponding to the standard basis vectors e_i is called a ***basis*** *for M.*

We say that M has ***rank*** *n if there exists a basis of order n. NB rank is not necessarily unique.*

Proposition 3.3.85
Let M be an (A, C)-bimodule then there is a canonical isomorphism of right C-modules

$$\begin{aligned}\mathrm{Hom}_A(A, M) \cong \quad & M \\ \theta \to \quad & \theta(1_A)\end{aligned}$$

This in particular includes the case $A = C$ is commutative.

Proposition 3.3.86
Let M be a free left A-module with basis $(m_i)_{i\in I}$ and N an (A, C)-bimodule then there is an isomorphism of right C-modules

$$\begin{aligned}\mathrm{Hom}_A(M, N) \cong \quad & \prod_{i\in I} N \\ \theta \to \quad & (\theta(m_i))_{i\in I}\end{aligned}$$

This in particular includes the case $A = C$ is commutative.

3.3.12 Free Modules

Definition 3.3.87 (Faithful Module)
We say an A-module M is ***faithful*** *if*

$$am = 0 \quad \forall m \in M \implies a = 0$$

Definition 3.3.88 (Linearly Independent, Spanning and Basis)
Let M be an A-module and $S \subset M$ a set. We say S is

- ***spanning*** *if $\langle S\rangle = M$*
- ***linearly independent*** *if for every finite subset $\{s_1, ..., s_n\} \subseteq S$ with s_i distinct we have*

$$\sum_{i=1}^{n} a_i s_i = 0 \implies a_i = 0 \quad 1 \le i \le n$$

- *a* ***basis*** *if it is both spanning and linearly independent*

Definition 3.3.89 (Finite Module)
An A-module M is ***finite*** *if there exists a finite spanning set.*

Definition 3.3.90 (Minimal spanning set)
Let M be an A-module. Then $S \subset M$ is a ***minimal spanning set*** *if it generates M and no proper subset does so.*

Definition 3.3.91 (Free Module)
Let M be an A-module. We say that M is a ***free module*** *over A if it has a basis.*

Proposition 3.3.92 (Free A-module is an external sum of A)
An A-module M is free if and only if it is isomorphic to $\bigoplus_{i\in I} A$ for some I. The isomorphism is given by

$$\sum_{i\in I} a_i m_i \to (a_i)_{i\in I}$$

Definition 3.3.93 (Internal Direct Sum Module)
An A-module M is an ***internal direct sum*** *of submodules $\{M_i\}_{i\in I}$ the canonical mapping of the external direct sum*

$$\bigoplus_{i\in I} M_i \to M$$

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is an isomorphism.

Proposition 3.3.94
Let M be an A-module and $\{M_i\}_{i\in I}$ a family of submodules. Then the following are equivalent

a) $\sum_{i\in I} M_i$ is the internal direct sum of the family $\{M_i\}_{i\in I}$
b) The relation $\sum_{i\in I} m_i = 0$ implies $m_i = 0$ for all $i \in I$
c) For any $i \in I$ we have $M_i \cap \left(\sum_{k\neq i} M_k\right) = \{0\}$

Proposition 3.3.95
*We say that M_1, M_2 are **supplementary** submodules of M if M is the internal direct sum of M_1 and M_2.*

*We say that M_1 is a **direct factor** of M if it is supplementary to another submodule.*

3.3.13 Exact Sequences

Definition 3.3.96 (Vector space)
*If k is a field and V a k-module, then we say V is a **vector space** over k.*

Remark 3.3.97
We will see that every vector space is free and every k-submodule is a direct factor.

Proposition 3.3.98 (Kernel)
*Let $\varphi : M \to N$ be an A-module homomorphism, then the **kernel** of φ*

$$\ker(\varphi) \coloneqq \{m \in M \mid \varphi(m) = 0\}$$

is an A-submodule of M. Observe φ is injective iff $\ker(\varphi) = 0$.

Proposition 3.3.99 (Image)
Let $\varphi : M \to N$ be an A-module homomorphism then the image

$$\mathrm{Im}(\varphi) = \{\varphi(m) \mid m \in M\}$$

is an A-submodule of N.

Definition 3.3.100 (Quotient Module)
*Let $N \subseteq M$ be an A-submodule then define the **quotient module** M/N to be the quotient group with an action of A given by*

$$a(m + N) = (am + N)$$

When $N \subseteq P \subseteq M$ is a sequence of submodules then define the A-submodule P/N of M/N by

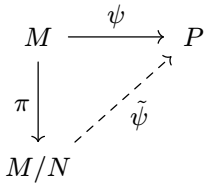
$$P/N \coloneqq \{p + N \mid p \in P\}$$

Proposition 3.3.101 (Quotient Module Properties)
Let $N \subseteq M$ be an A-submodule then there is a canonical surjective morphism

$$\pi : M \to M/N$$

with the following properties

a) $\pi(m) = m + N$
b) $\ker(\pi) = N$
c) Every homomorphism $\psi : M \to P$ such that $N \subseteq \ker(\psi)$, factors uniquely through π



Furthermore there is a bijection of A-submodules

$$\{P' \subseteq M/N\} \leftrightarrow \{P \mid N \subseteq P \subseteq M\}$$

given by $P' = P/N$. In the situation above $\ker(\tilde{\psi}) = \ker(\psi)/N$. In particular if $\ker(\psi) = N$ then $\tilde{\psi}$ is injective.

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Corollary 3.3.102
Let $\psi : M \to N$ be an A-module homomorphism, then this induces an isomorphism

$$M/\ker(\psi) \cong \mathrm{Im}(\psi)$$

Definition 3.3.103 (Exact Sequence)
*Let $N \xrightarrow{\varphi} M \xrightarrow{\psi} P$ be an sequence of A-module homomorphisms. We say it is **exact** if*

$$\mathrm{Im}(\varphi) = \ker(\psi)$$

It is equivalent to the following two conditions

a) $\psi \circ \varphi = 0$
b) $\psi(m) = 0 \implies m = \varphi(n)$ *for some* $n \in N$.

An exact sequence of the form

$$0 \to N \to M \to P \to 0$$

*is said to be **short-exact**.*

Remark 3.3.104
There are a few trivial observations

- $0 \to M \to N$ *is exact if and only if the map* $M \to N$ *is injective*
- $M \to N \to 0$ *is exact if and only if the map* $M \to N$ *is surjective.*

Proposition 3.3.105 (Isomorphism induced by short-exact sequence)
Let $N \subseteq M$ be a A-submodule then there is a canonical short-exact sequence

$$0 \to N \to M \to M/N \to 0$$

Conversely suppose we have a short exact sequence

$$0 \to N \xrightarrow{i} M \xrightarrow{\pi} P \to 0$$

then this induces an isomorphism

$$M/i(N) \cong P$$

If N is a submodule of M then we would simply write $M/N \cong P$.

Proposition 3.3.106 (Third Isomorphism Theorem)
Let $N \subseteq N' \subseteq M$ be a chain of modules then there is a short-exact sequence

$$0 \to N'/N \to M/N \to M/N' \to 0$$

which then induces an isomorphism

$$(M/N)/(N'/N) \cong M/N'$$

Proposition 3.3.107 (Product of ideal and quotient module)
Let N be a submodule of M and $\mathfrak{a} \lhd A$ an ideal. Then

$$\mathfrak{a}(M/N) = (N + \mathfrak{a}M)/N$$

Proposition 3.3.108 (Induced module)
Let M be an A-module and $\mathfrak{a}$ an ideal such that $\mathfrak{a}M = 0$, then M is naturally an $A/\mathfrak{a}$-module with action given by

$$\overline{a} \cdot m := a \cdot m$$

3.3.14 Dual Module

Proposition 3.3.109 (Dual Module)
Let M be a left (resp. right) A-module. Then the set

$$M^\vee := \mathrm{Hom}_A(M, A)$$

is canonically a right *(resp.* left*) A-module.*

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If M is a finite free left (resp. right) A-module with basis $\{v_1, ..., v_n\}$ then $M^\vee$ is a finite free right *(resp.* left*) A-module with basis $\{v_1^\vee, ..., v_n^\vee\}$ where these are the unique homomorphisms satisfying*

$$v_i^\vee(v_j) = \delta_{ij}$$

Moreover every basis of $M^\vee$ is of this form.

Proposition 3.3.110 (Hom-Set is free)
Let M be a finite free left A-module with basis $\{v_1, ..., v_n\}$ and N a (A, B)-bimodule. Then there is an isomorphism of right B-modules

$$\begin{aligned} \mathrm{Hom}_A(M, N) \cong & \bigoplus_{i=1}^{n} N \\ \theta \to & (\theta(v_i))_{i \in I} \end{aligned}$$

If A is commutative and N is a finite-free A-module with basis $\{w_1, ..., w_m\}$ then there is further an isomorphism of A-modules

$$\begin{aligned} \mathrm{Hom}_A(M, N) \cong & \bigoplus_{i,j=1}^{n,m} A \\ \theta \to & \left(w_j^\vee(\theta(v_i))\right)_{i,j} \end{aligned}$$

where $w_1^\vee, ..., w_m^\vee$ is the dual basis for $N^\vee$. In particular $\mathrm{Hom}_A(M, N)$ is a finite-free A-module with basis

$$\left\{w_j v_i^\vee\right\}_{i,j}$$

Definition 3.3.111 (Dual Functor)
Let A be a commutative ring and $\varphi : M \to N$ an A-module homomorphism. Define the dual homomorphism $\varphi^\vee : N^\vee \to M^\vee$ by

$$\varphi^\vee(\psi) := \psi \circ \varphi$$

This determines a contravariant functor

$$(-)^\vee : A\textbf{-Mod} \to A\textbf{-Mod}$$

Corollary 3.3.112 (Double Dual Natural Isomorphism)
Let A be a commutative ring and M a finite free A-module then the canonical A-module homomorphism

$$\begin{aligned} \eta : M \longrightarrow & \quad M^{\vee\vee} \\ x \to & \quad (\varphi \to \varphi(x)) \end{aligned}$$

is an isomorphism, which is natural in M.

Corollary 3.3.113
The contravariant functor $(-)^\vee : \mathbf{FiniteFreeMod}_A \to \mathbf{FiniteFreeMod}_A$ is an equivalence of categories and therefore full and faithful.

Proof. Use the dual isomorphism η together with (2.6.29) and (2.6.28). ∎

3.3.15 Matrices

For this section we assume that A is a commutative ring. In this context $A^n := A \times ... \times A$ is a finite free module with basis $e_1, ..., e_n$. Matrices are concrete realisations of linear maps of finite free modules.

Proposition 3.3.114
Let M, N be finite free A-modules with ordered bases $\mathcal{B} := \{v_1, ..., v_n\}$, $\mathcal{B}' := \{w_1, ..., w_m\}$ respectively. Let $\varphi : M \to N$ be an A-linear map and define $[\varphi]$ to be the unique $m \times n$ matrix satisfying the following condition

$$\varphi(v_i) = \sum_{j=1}^{m} [\varphi]_{ji} w_j \quad \forall i = 1...n$$

*(**notice sum over the first index**). Then it is also the unique matrix ensuring that the following diagram commutes*

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$$\begin{array}{ccc} M & \xrightarrow{\varphi} & N \\ \uparrow \sim & & \uparrow \sim \\ A^n & \xrightarrow{x \to [\varphi]x} & A^m \end{array}$$

where the bottom arrow is given by usual multiplication of a matrix by a column vector

Proposition 3.3.115 (Matrices as linear maps)
Let M, N be free A-modules with ordered bases $\mathcal{B} := \{v_1, ..., v_n\}$, $\mathcal{B}' := \{w_1, ..., w_m\}$ respectively. Then there are mutually inverse isomorphisms of A-modules

$$\begin{aligned} \mathrm{Mat}_{m\times n}(A) &\longleftrightarrow \mathrm{Hom}_A(M, N) \\ E &\to \hat{E} \\ [\varphi]^{\mathcal{B}}_{\mathcal{B}'} &\leftarrow \varphi \end{aligned}$$

where

$$\begin{aligned} \hat{E}\left(\sum_{i=1}^{n} \lambda_i v_i\right) &:= \sum_{j=1}^{m}\left(\sum_{i=1}^{n} E_{ji}\lambda_i\right) w_j \\ \varphi(v_i) &= \sum_{j=1}^{m} [\varphi]_{ji} w_j \end{aligned}$$

If we further consider a free A-module P with ordered bases $\mathcal{B}'' = \{u_1, ..., u_p\}$ then

$$\begin{aligned} \hat{E} \circ \hat{F} &= \widehat{EF} \\ [\psi \circ \varphi]^{\mathcal{B}}_{\mathcal{B}''} &= [\psi]^{\mathcal{B}'}_{\mathcal{B}''}[\varphi]^{\mathcal{B}}_{\mathcal{B}'} \end{aligned}$$

Observe that

$$[1_M]^{\mathcal{B}}_{\mathcal{B}} = I_n$$

Furthermore there is an isomorphism of A-algebras

$$\begin{aligned} \mathrm{End}_A(M) &\longleftrightarrow \mathrm{Mat}_{n,n}(A) \\ \varphi &\to \left(v_i^\vee(\varphi(v_j))\right)_{ij} \\ \sum_{ij} E_{ij} v_i v_j^\vee &\leftarrow E \end{aligned}$$

Corollary 3.3.116
Matrix multiplication is associative. In particular

$$(EF)v = E(Fv)$$

Proof. We may consider the free A-modules A^n, A^m and A^p with canonical bases. The result follows because function composition is associative and $\hat{\cdot}$ is injective. ∎

Corollary 3.3.117
There is an isomorphism of A-modules

$$\begin{aligned} \mathrm{Mat}_{m\times n}(A) &\longleftrightarrow \mathrm{Hom}_A(A^n, A^m) \\ E &\to (v \to Ev) \end{aligned}$$

and further an isomorphism of A-algebras

$$\mathrm{Mat}_{n,n}(A) \longleftrightarrow \mathrm{End}_A(A^n)$$

Corollary 3.3.118
Let M be a finite free A-module with bases $\mathcal{B}, \mathcal{B}'$ and $\varphi \in \mathrm{End}_A(M)$. Then φ is an isomorphism if and only if $[\varphi]^{\mathcal{B}}_{\mathcal{B}'} \in \mathrm{Mat}_{n\times n}(A)$ is invertible.

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Corollary 3.3.119 (Change of basis)
Let M be a finite free A-module and $\mathcal{B}, \mathcal{B}'$ bases then

$$[1_M]^{\mathcal{B}}_{\mathcal{B}'} = \left([1_M]^{\mathcal{B}'}_{\mathcal{B}}\right)^{-1}$$

and

$$[\varphi]^{\mathcal{B}'}_{\mathcal{B}'} = P[\varphi]^{\mathcal{B}}_{\mathcal{B}}P^{-1}$$

where

$$P := [1_M]^{\mathcal{B}}_{\mathcal{B}'}$$

is invertible.

Definition 3.3.120 (Transpose)
*Let E be an $m \times n$ matrix in A, then define the **transpose** of E to be the $n \times m$ matrix E^t where*

$$\left(E^t\right)_{ij} := E_{ji}$$

Proposition 3.3.121
Let M, N be finite-free A-modules with bases $\mathcal{B} = \{v_1, ..., v_n\}$ and $\mathcal{B}' = \{w_1, ..., w_m\}$. Let $\varphi : M \to N$ be an A-module homomorphism and $\varphi^\vee : N^\vee \to M^\vee$ the dual homomorphism then

$$[\varphi^\vee]^{\mathcal{B}'^\vee}_{\mathcal{B}^\vee} = \left([\varphi]^{\mathcal{B}}_{\mathcal{B}'}\right)^t$$

Similarly if E is an $m \times n$ matrix over A then

$$\hat{E}^\vee = \widehat{E^t}$$

where the right hand side is understood to be with respect to the dual bases.

Corollary 3.3.122
Let E, F be matrices then

$$(EF)^t = (FE)^t$$

3.3.16 Vector Spaces

Definition 3.3.123 (Vector Space)
*A **vector space** V over k is simply a k-module.*

*A k-submodule is referred to as a **subspace***

*A k-module homomorphism is referred to as a **linear map***

A vector space is finite-dimensional if it is finite as a k-module.

The main result on vector spaces is that bases exist and all have the same cardinality. Recall that $\langle\cdot\rangle$ is a closure operator. We show that $(V, \langle\cdot\rangle)$ determines a matroid so that we may appeal to results in the Matroids section. First we need to show that the notions of independence coincide

Proposition 3.3.124 (Equivalent definitions of linear independence)
Let V be a vector space and $S \subset V$. Then the following are equivalent

a) S is linearly independent
b) No proper subset $S' \subset S$ satisfies $\langle S' \rangle = \langle S \rangle$
*c) **Matroid Independence** $x \in S \implies x \notin \langle S \setminus \{x\} \rangle$*

Further S is independent if and only if every finite subset of S is.

Proof. b) $\iff$ c) This is b).

a) $\implies$ c). If $x \in \langle S \setminus \{x\} \rangle$ then it's clear that S is not linearly independent.

c) $\implies$ a). Suppose we have a linear relationship

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$$0 = \sum_{i=1}^{n} \lambda_i v_i \quad v_i \in S$$

By renumbering assume that $\lambda_1 \neq 0$, then rearrange to show $v_1 \in \langle S \setminus \{v_1\} \rangle$, contradicting the hypothesis. ∎

We show that $(V, \langle \cdot \rangle)$ satisfies the Exchange Property and therefore constitutes a matroid.

Proposition 3.3.125 (Exchange Property)
Let V be a vector space and $S \subset V$. Then

$$y \in \langle S \cup \{x\} \rangle \setminus \langle S \rangle \implies x \in \langle S \cup \{y\} \rangle$$

Proof. Suppose y is as given, then

$$y = \lambda x + \sum_i \lambda_i s_i \quad s_i \in S$$

The case $\lambda = 0$ is obvious. Otherwise we may rearrange to find

$$x = \lambda^{-1} y - \sum_i \lambda^{-1} \lambda_i s_i$$

whence $x \in \langle S \cup \{y\} \rangle$. ∎

Therefore we have the following

Proposition 3.3.126 (Vector Spaces are Free)
Every vector space has a basis, and in the finite-dimensional case every basis is finite of the same size. We denote this by $\dim_k V$.

More generally every linearly independent set is contained in a basis (so has order at most $\dim_k V$) and every spanning set contains a basis (so has order at least $\dim_k V$)

Proof. Follows from (2.3.7) and (2.3.12). The final statement follows from (2.3.13). ∎

Proposition 3.3.127 (Basis Criteria)
Let V be a vector space with $n = \dim_k V$ and $\mathcal{B} \subseteq V$. Then TFAE

a) $\mathcal{B}$ is a basis
b) $\mathcal{B}$ is linearly independent and $\#\mathcal{B} \geq \dim_k V$
c) $\mathcal{B}$ is spanning and $\#\mathcal{B} \leq \dim_k V$

Proof. The equivalence of a), b) and c) follows from (2.3.14). ∎

Proposition 3.3.128
A vector space $V = \{0\}$ if and only if $\dim_k V = 0$

Proposition 3.3.129 (Image of a basis)
Let $\varphi : V \to W$ be a linear map

a) If S is linearly-independent and φ is injective, then $\varphi(S)$ is linearly-independent
b) If Γ is spanning then (φ is surjective $\iff$ $\varphi(\Gamma)$ is spanning)
c) If $\mathcal{B}$ is a basis then (φ is an isomorphism $\iff$ $\varphi(\mathcal{B})$ is a basis and φ injective on $\mathcal{B}$)

Proof.

a) Suppose $\sum_i \lambda_i \varphi(s_i) = 0 \implies \varphi\left(\sum_i \lambda_i s_i\right) = 0$. As φ is injective this implies $\sum_i \lambda_i s_i = 0 \implies \lambda_i = 0$.
b) If φ is surjective then for $w \in W$ we have $\varphi(v) = w$ for some $v \in V$. By hypothesis $v = \sum_i \lambda_i \gamma_i$ and $w = \sum_i \varphi(\lambda_i)$. Conversely given $w \in W$ by hypothesis $w = \sum_i \lambda_i \varphi(\gamma_i) = \varphi\left(\sum_i \lambda_i \gamma_i\right)$ and φ is surjective as required.
c) Suppose φ is isomorphism, then it's surely injective on $\mathcal{B}$ and by a),b) $\varphi(\mathcal{B})$ is a basis. Conversely if $\varphi(\mathcal{B}) =: \mathcal{B}'$ is a basis then by b) φ is surjective. Suppose $\varphi(v) = 0$. Then by hypothesis $v = \sum_i \lambda_i v_i$ for $v_i \in \mathcal{B}$ and $0 = \varphi(v) = \sum_i \lambda_i \varphi(v_i)$. By hypothesis $\varphi(v_i)$ are distinct elements of the basis $\mathcal{B}'$ and therefore $\lambda_i = 0$ and $v = 0$. Therefore φ is injective and hence bijective.

∎

Corollary 3.3.130 (Dimension is an invariant)
Dimension is preserved under isomorphism. More generally for $\varphi : V \to W$ we have

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$$\varphi \text{ injective} \implies \dim_k V \leq \dim_k W$$

$$\varphi \text{ surjective} \implies \dim_k V \geq \dim_k W$$

Proposition 3.3.131
Let $W \subseteq V$ be finite-dimensional vector spaces then the dimension of the quotient module satisfies

$$\dim_k V/W = \dim_k V - \dim_k W$$

Proof. Let $\{v_1, ..., v_m\}$ be a basis of W, then there exists a basis $\{v_1, ..., v_m, v_{m+1}, ..., v_n\}$ of V containing the first by (3.3.126). We claim that

$$\{[v_{m+1}], ..., [v_n]\}$$

is a basis for V/W, and the result follows. For given $[v] \in V/W$ then

$$v = \sum_{i=1}^{n} \lambda_i v_i$$

since the basis is spanning. We have

$$v - \sum_{i=m+1}^{n} \lambda_i v_i \in W$$

therefore

$$[v] = \left[\sum_{i=m+1}^{n} \lambda_i v_i\right] = \sum_{i=m+1}^{n} \lambda_i [v_i]$$

and the given set is spanning. Similarly suppose

$$\sum_{i=m+1}^{n} \lambda_i [v_i] = 0$$

then by definition $\sum_{i=m+1}^{n} \lambda_i v_i \in W$. Therefore

$$\sum_{i=m+1}^{n} \lambda_i v_i = \sum_{i=1}^{m} \lambda_i v_i$$

and since v_i are linearly independent we must have $\lambda_i = 0$. ∎

Proposition 3.3.132 (Injective Criteria)
Let $\varphi : V \to W$ be a linear map then

$$\varphi \text{ injective} \iff \ker(\varphi) = \{0\} \iff \dim_k \ker(\varphi) = 0$$

Proof. Note for any linear map φ we have $\varphi(0) = 0$. Therefore φ injective clearly shows $\ker(\varphi) = \{0\}$. Conversely suppose $\ker(\varphi) = 0$ and $\varphi(v) = \varphi(w)$. Then $\varphi(v - w) = 0 \implies v - w = 0 \implies v = w$ as required. ∎

Definition 3.3.133 (Rank)
Let $\varphi : V \to W$ be a linear map then define

$$\operatorname{rank}_k(\varphi) := \dim_k(\operatorname{Im}(\varphi))$$

Proposition 3.3.134 (Surjective Criteria)
Let $\varphi : V \to W$ be a linear map with W finite-dimensional then

$$\varphi \text{ surjective} \iff \operatorname{rank}_k(\varphi) = \dim_k W$$

Proof. This follows directly from (2.3.15). ∎

Proposition 3.3.135 (Isomorphism Theorem / Rank-Nullity)
Let $\varphi : V \to W$ be a linear map then this induces an isomorphism

$$V/\ker(\varphi) \xrightarrow{\sim} \operatorname{Im}(\varphi)$$

in particular when V is finite-dimensional

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$$\dim_k V = \dim_k \ker(\varphi) + \operatorname{rank}_k(\varphi)$$

Corollary 3.3.136 (Isomorphism Criteria)
Let V, W vector spaces with W finite-dimensional. A linear map $\varphi : V \to W$ is an isomorphism if and only if any two *of the following are satisfied*

a) $\dim_k \ker(\varphi) = 0 \iff \varphi$ *injective*
b) $\dim_k V = \dim_k W$
c) $\operatorname{rank}_k(\varphi) = \dim_k W \iff \varphi$ *surjective*

Proof. The rank-nullity equation ensures that if any two hold the third is automatically satisfied. In this case φ is isomorphism as required. ∎

Corollary 3.3.137 (Endomorphism Isomorphism Criteria)
Let V be a finite-dimensional vector space and $\varphi : V \to V$ then TFAE

a) φ *is injective*
b) φ *is surjective*
c) φ *is an isomorphism*

Proof. For the equivalence of a), b) and c) we may use the previous result with $W = V$ and note $\dim_k W = \dim_k V$ is automatically satisfied. ∎

Proposition 3.3.138 (Internal Direct Sum)
Let U_1, U_2 be two subspaces of V then TFAE

a) $U_1 \cap U_2 = \{0\}$ *and* $V = U_1 + U_2$
b) Every $v \in V$ *may be written uniquely as* $u_1 + u_2$ *for* $u_i \in U_i$.

and we say $V = U_1 \oplus U_2$ is an ***internal direct sum*** *and U_2 is a* ***supplementary subspace*** *for U_1.*

Proposition 3.3.139 (Subspaces are direct factors)
Every subspace U has a supplementary subspace U' such that

$$V = U \oplus U'$$

Proof. Let $\mathcal{B}_1$ be a basis for U and extend to a basis $\mathcal{B}$ and define $\mathcal{B}_2 := \mathcal{B} \setminus \mathcal{B}_1$. Then it's easy to show that $U' = \langle \mathcal{B}_2 \rangle$ is a supplementary subspace. ∎

Proposition 3.3.140 (Dimension formula for direct sums)
Suppose $V = U_1 \oplus U_2$, $\mathcal{B}_1$ is a basis for U_1 and $\mathcal{B}_2$ is a basis for U_2. Then $\mathcal{B}_1 \cap \mathcal{B}_2 = \emptyset$ and $\mathcal{B}_1 \cup \mathcal{B}_2$ is a basis for V. In particular

$$\dim_k V = \dim_k U_1 + \dim_k U_2$$

3.3.16.1 Dual Space

Definition 3.3.141 (Dual Space)
Let V be a k-vector space and define the ***dual space*** *to be*

$$V^\vee := \operatorname{Hom}_k(V, k)$$

This is an abelian group and even a k-vector space under the obvious operations. The construction $V \to V^\vee$ determines a contravariant functor

$$(-)^\vee : \mathbf{Vect}_k \to \mathbf{Vect}_k$$

Definition 3.3.142 (Annihilator)
Let V be a vector space and $U \subseteq V$ a subspace. Define the ***annihilator*** *of U by*

$$U^\circ = \{\theta \in V^\vee \mid \theta(u) = 0 \quad \forall u \in U\}$$

This is a linear subspace of $V^\vee$.

Proposition 3.3.143 (Dimension formula for annihilators)
There is a canonical isomorphism by restriction

$$V^\vee / U^\circ \to U^\vee$$

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In particular when V is a finite-dimensional vector space then

$$\dim_k V = \dim_k U + \dim_k U^\circ$$

Proof. Let W be a supplementary subspace and consider the morphism $V = U \oplus W \xrightarrow{\pi_U} U$. Then $(\theta \circ \pi_U)|_U = \theta$ so the restriction map is surjective. Clearly the kernel is U°. The dimension formula follows from (3.3.135) and (3.3.109). ∎

Corollary 3.3.144 (Dual rank = rank)
Let $\varphi : V \to W$ be a linear map and $\varphi^\vee : W^\vee \to V^\vee$ then

$$\begin{aligned} \ker(\varphi^\vee) &= \operatorname{Im}(\varphi)^\circ \\ \operatorname{Im}(\varphi^\vee) &\subseteq \ker(\varphi)^\circ \end{aligned}$$

In the finite-dimensional case $\operatorname{Im}(\varphi^\vee) = \ker(\varphi)^\circ$ and

$$\begin{aligned} \dim_k \ker(\varphi^\vee) &= \dim_k W - \operatorname{rank}_k(\varphi) \\ \operatorname{rank}_k(\varphi^\vee) &= \operatorname{rank}_k(\varphi) \end{aligned}$$

Proof. Note $\ker(\varphi^\vee) = \operatorname{Im}(\varphi)^\circ$ and $\operatorname{Im}(\varphi^\vee) \subseteq \ker(\varphi)^\circ$ by the definitions.

Consider the finite-dimensional case. By (3.3.143)

$$\dim_k \ker(\varphi^\vee) = \dim_k \operatorname{Im}(\varphi)^\circ = \dim_k W - \operatorname{rank}_k(\varphi)$$

By rank-nullity applied to $\varphi^\vee$ and $\dim_k W = \dim_k W^\vee$ we deduce

$$\operatorname{rank}_k(\varphi^\vee) = \operatorname{rank}_k(\varphi).$$

By (3.3.143) and rank-nullity applied to φ

$$\dim_k \ker(\varphi)^\circ = \dim_k V - \dim_k \ker(\varphi) = \operatorname{rank}_k(\varphi).$$

Finally by (2.3.15) $\operatorname{Im}(\varphi^\vee) = \ker(\varphi)^\circ$. ∎

From this it follows that taking duals reflects and preserve isomorphisms

Corollary 3.3.145 ($(-)^\vee$ reflects isomorphisms)
Let $\varphi : V \to W$ be a linear map of finite-dimensional spaces then

a) φ is injective if and only if $\varphi^\vee$ is surjective
b) φ is surjective if and only if $\varphi^\vee$ is injective
c) φ is iso if and only if $\varphi^\vee$ is iso

Proof. Note by (3.3.144) we have $\operatorname{rank}_k(\varphi) = \operatorname{rank}_k(\varphi^\vee)$ and by (3.3.109) $\dim_k V^\vee = \dim_k V$.

φ is surjective $\iff \operatorname{rank}_k(\varphi) = \dim_k W = \operatorname{rank}_k(\varphi^\vee) \overset{(3.3.135)}{\iff} \dim_k \ker(\varphi^\vee) = 0 \iff \ker(\varphi^\vee) = \{0\}$

φ is injective $\iff \dim_k \ker(\varphi) = 0 \overset{(3.3.135)}{\iff} \operatorname{rank}_k(\varphi) = \dim_k V \iff \dim_k V^\vee = \operatorname{rank}_k(\varphi^\vee) \iff \varphi^\vee$ is surjective.

The last point may be deduced from the first two, or the fact that $(-)^\vee$ is full and faithful (3.3.113) and category-theoretic result (2.6.42). ∎

3.3.16.2 Bilinear Pairings

Definition 3.3.146 (Bilinear maps)
Let V, W be vector spaces a bilinear map ψ is a map

$$\psi : V \times W \to k$$

which is k-linear in each variable. We denote the set of bilinear maps as

$$\operatorname{Bilin}_k(V, W)$$

Proposition 3.3.147 (Matrix Representation)
Let V, W be finite-dimensional vector spaces with bases $\{v_1, ..., v_n\}$ and $\{w_1, ..., w_m\}$ then there is a canonical isomorphism

$$\begin{aligned} \operatorname{Bilin}_k(V, W) &\xrightarrow{\sim} \operatorname{Mat}_{n,m}(k) \\ \psi &\to \left(\psi(v_i, w_j)\right)_{ij} \end{aligned}$$

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In particular a bilinear map ψ is determined uniquely by the values $\psi(v_i, w_j)$.

Proposition 3.3.148 (Dual maps)
Let V and W be vector spaces, then there is a natural bijection

$$\begin{array}{ccccc} \mathrm{Mor}_k(V, W^\vee) & \leftrightarrow & \mathrm{Bilin}_k(V, W) & \leftrightarrow & \mathrm{Mor}_k(W, V^\vee) \\ \psi_L & \leftarrow & \psi & & \\ & & \psi & & \to \psi_R \end{array}$$

where

$$\psi_L(v)(w) = \psi(v, w) = \psi_R(w)(v)$$

When V, W are finite-dimensional then ψ_L is an isomorphism if and only if ψ_R is an isomorphism. In this case we say ψ is a perfect pairing. More generally

$$\mathrm{rank}_k(\psi_L) = \mathrm{rank}_k(\psi_R)$$

Proof. The bijections stated are obvious. One may show that $\psi_L = \psi_R^\vee \circ \eta_V$ where $\eta_V : V \to V^{\vee\vee}$ is the dual isomorphism. Therefore ψ_L is an isomorphism if and only if $\psi_R^\vee$ is an isomorphism, and by (3.3.145) if and only if ψ_R is an isomorphism. Since η_V is surjective we have $\mathrm{rank}_k(\psi_L) = \mathrm{rank}_k(\psi_R^\vee) = \mathrm{rank}_k(\psi_R)$, by (3.3.144). ∎

Proposition 3.3.149
Let $\psi : V \times V \to k$ be a bilinear map. Then the following are equivalent

a) ψ is a perfect pairing
b) ψ_L is an isomorphism
c) ψ_R is an isomorphism
d) ψ_L is injective
e) ψ_R is injective
f) ψ_L is surjective
g) ψ_R is surjective.

Proof. We may use the fact $\dim_k V = \dim_k V^\vee$ and (3.3.136) to show b) ⟺ d) ⟺ f) and c) ⟺ e) ⟺ g). We've already shown that a) ⟺ b) ⟺ c). ∎

Definition 3.3.150 (Orthogonal Complement)
Let $\psi : V \times W \to k$ be a perfect pairing of finite-dimensional vector spaces. Suppose $U \subset V$ is a subspace then define the ***orthogonal complement***

$$U^\perp := \{w \in W \mid \psi(v, w) = 0 \quad \forall v \in U\}$$

Proposition 3.3.151
Let $\psi : V \times W \to k$ be a perfect pairing of finite-dimensional vector spaces and $U \subset V$ a subspace. Then

$$\dim_k U + \dim_k U^\perp = \dim_k V$$

Indeed ψ_R induces an isomorphism $U^\perp \to U^\circ$.

Proof. We claim that $\psi_R(U^\perp) = U^\circ$. For if $w \in U^\perp$ then $\psi_R(w)(v) = \psi(w, v) = 0$ for all $v \in U$, and so $\psi_R(w) \in U^\circ$. Conversely given $\theta \in U^\circ$, as ψ_R is surjective, there is $w \in W$ such that $\psi_R(w) = \theta$. By definition $w \in U^\perp$ as required.

As ψ_R is injective then $\dim_k U^\perp \overset{(3.3.130)}{=} \dim_k U^\circ \overset{(3.3.143)}{=} \dim_k V/U = \dim_k V - \dim_k U$. ∎

Remark 3.3.152
In the case $V = W$, then it's not necessarily true that $U \cap U^\perp = \{0\}$, and so $U^\perp$ is not necessarily a complementary subspace.

The classic example is the perfect pairing on $\mathbb{R}^n$ induced by vDv^T for a real diagonal matrix D. Then it's true in general if and only if D is positive-definite.

Proposition 3.3.153 (Quotients are dual to subspaces)
Let $\psi : V \times W \to k$ be a perfect pairing of finite-dimensional vector spaces. Suppose $U \subset V$ is a subspace, then there is a canonical perfect pairing

$$\psi' : V/U \times U^\perp \to k$$

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given by

$$\psi'(v+U,w)=\psi(v,w)$$

Proof. The given map is well defined, for suppose $v_1+U=v_2+U$ then $v_1-v_2\in U\implies\psi(v_1-v_2,w)=0\quad\forall w\in U^\perp\implies\psi(v_1,w)=\psi(v_2,w)$ as required. It's clearly k-bilinear.

It's clear that ψ'_R is injective, because $\psi'_R(w)=0_{V/U}\implies\psi_R(w)=0_V\implies w=0$.

By the previous Proposition $\dim_k U^\perp=\dim_k V/U$. Therefore by (3.3.136) ψ'_R is an isomorphism and ψ' is perfect.∎

3.3.17 Inverse Limits

We record the basic construction of an inverse limit of a tower of rings, modules, or algebras, as a concrete model for the categorical inverse limit (2.6.130), together with its functoriality and its compatibility with products; these are used repeatedly to build completions later on.

Definition 3.3.154 (Inverse Limit of a Tower)
Let $\mathcal{C}$ be one of **Ring**, *A-***Mod** *(for a fixed ring A), or Alg_A (for a fixed ring A, so in particular* $\mathbf{Alg}_k$*) – each an algebraic category (2.6.185) (via the usual axioms; (2.6.186), (2.6.187), (2.6.188) respectively). Let $\cdots\twoheadrightarrow M_2\twoheadrightarrow M_1\twoheadrightarrow M_0$ be a sequence of morphisms $\rho_n:M_n\to M_{n-1}$ of $\mathcal{C}$ $(n\geq 1)$. Define*

$$\varprojlim_n M_n:=\left\{(m_n)_{n\geq0}\in\prod_{n\geq0}M_n\;\middle|\;\rho_n(m_n)=m_{n-1}\text{ for every }n\geq1\right\},$$

a sub-object of $\prod_n M_n$ in $\mathcal{C}$ (a subring, submodule, or subalgebra respectively, the ambient operations acting componentwise), with the evident projections $\lambda_n:\varprojlim_m M_m\to M_n$. Writing $\rho_{n,n}:=1_{M_n}$ and $\rho_{n,m}:=\rho_{m+1}\circ\cdots\circ\rho_n:M_n\to M_m$ for $n>m$ (composites of consecutive ρ_k's), every $(m_n)_{n\geq0}\in\varprojlim_n M_n$ satisfies the ***repeated compatibility***

$$\rho_{n,m}(m_n)=m_m\quad\text{for all }n\geq m,$$

not just for consecutive n,m: by induction on $n-m$, composing the defining relations $\rho_k(m_k)=m_{k-1}$ for $m<k\leq n$.

Lemma 3.3.155 (This Realises the Categorical Inverse Limit)
In the setting of (3.3.154), regard $I:=\mathbb{N}$ (usual order) as a directed preorder and the tower as an inverse system $F:I^{op}\to\mathcal{C}$, $F(n):=M_n$, with restriction maps $\rho_{n,m}$ as in (3.3.154). Then $\left(\varprojlim_n M_n,\{\lambda_n\}\right)$ is an inverse limit for F in the sense of (2.6.130).

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

$\mathcal{C}$ is an algebraic category (2.6.185), in each of the three cases (as recalled in (3.3.154)). By (2.6.192), $\varprojlim_n F(n)$ exists in $\mathcal{C}$, with underlying set $\left\{(m_n)_n\in\prod_n M_n:\rho_{n,m}(m_n)=m_m\text{ for all }n\geq m\right\}$ and the ambient operations acting coordinatewise; by the repeated compatibility (3.3.154), this defining condition holds for all $n\geq m$ as soon as it holds for consecutive $n=m+1$, so this is exactly the sub-object $\varprojlim_n M_n$ of (3.3.154), with the same (componentwise) structure and projections. So $\left(\varprojlim_n M_n,\{\lambda_n\}\right)$ is this categorical inverse limit. ∎

Corollary 3.3.156 (Functoriality of the Inverse Limit)
Let $\mathcal{C}$ be as in (3.3.154), $(A_n,\rho_n)_{n\geq0}$ and $(B_n,\tau_n)_{n\geq0}$ towers in $\mathcal{C}$, and for each $n\geq0$ let $\sigma_n:A_n\to B_n$ be a morphism of $\mathcal{C}$ such that, for every $n\geq1$, the square

$$\begin{array}{ccc} A_n & \xrightarrow{\sigma_n} & B_n \\ {\scriptstyle\rho_n}\downarrow & & \downarrow{\scriptstyle\tau_n} \\ A_{n-1} & \xrightarrow[\sigma_{n-1}]{} & B_{n-1} \end{array}$$

commutes, i.e. $\tau_n\circ\sigma_n=\sigma_{n-1}\circ\rho_n$. Then

$$\varprojlim\sigma:\varprojlim_n A_n\longrightarrow\varprojlim_n B_n,\quad(a_n)_n\mapsto(\sigma_n(a_n))_n$$

is a well-defined morphism of $\mathcal{C}$; and if every σ_n is an isomorphism, so is $\lim_{\leftarrow}\sigma$.

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Proof. By (3.3.155), $\lim_{\leftarrow n} A_n$ and $\lim_{\leftarrow n} B_n$ (with their projections) are inverse limits, in the sense of (2.6.130), of the inverse systems $F, G : \mathbb{N}^{\mathrm{op}} \to \mathcal{C}$ built from the two towers; the commuting squares above are exactly the naturality hypothesis of (2.6.133) for the family $\sigma = (\sigma_n)_n$ (composing squares for consecutive n gives it for the composite restriction maps $\rho_{n,m}$, $\tau_{n,m}$, $n > m$, as needed there). That corollary gives the induced morphism $\lim_{\leftarrow n} \sigma_n$, agreeing on each coordinate with σ_n by construction of the induced map on limits-as-sets (2.6.131) – that is, the map $(a_n)_n \to (\sigma_n(a_n))_n$ above – and an isomorphism whenever every σ_n is. ∎

Proposition 3.3.157 (Inverse Limits Commute with Products)
Let $\mathcal{C}$ be as in (3.3.154), K a set, and for each $k \in K$ a tower $(M_{k,n}, \rho_{k,n})_{n\geq 0}$ in $\mathcal{C}$. Then $\left(\prod_{k\in K} M_{k,n}\right)_{n\geq 0}$, with restriction maps $\prod_{k\in K} \rho_{k,n} : \prod_k M_{k,n} \to \prod_k M_{k,n-1}$, is again a tower in $\mathcal{C}$, and

$$\Phi : \prod_{k\in K} \lim_{\leftarrow n} M_{k,n} \longrightarrow \lim_{\leftarrow n} \left(\prod_{k\in K} M_{k,n}\right), \quad \Phi\big((x_k)_{k\in K}\big) := \Big(\big(x_{k,n}\big)_{k\in K}\Big)_{n\geq 0} \quad \big(x_k = (x_{k,n})_{n\geq 0}\big)$$

is a well-defined isomorphism of $\mathcal{C}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Write $F_k : \mathbb{N}^{\mathrm{op}} \to \mathcal{C}$ for the inverse system of the tower $(M_{k,n})_n$, so $\left(\lim_{\leftarrow n} M_{k,n}, \{\lambda_{k,n}\}_n\right)$ realises $\lim_{\leftarrow} F_k$ (3.3.155), and likewise $\left(\lim_{\leftarrow n}\left(\prod_k M_{k,n}\right), \{\Lambda_n\}_n\right)$ realises $\lim_{\leftarrow}\left(\prod_k F_k\right)$, where Λ_n is projection onto the n-th coordinate. Since $\prod_{k\in K} \lim_{\leftarrow n} M_{k,n}$ (with the evident coordinate projections p_k) is the categorical product of the $\lim_{\leftarrow n} M_{k,n}$ in $\mathcal{C}$, (2.6.134) gives that $\lim_{\leftarrow n}\left(\prod_k M_{k,n}\right)$ exists (as it already does, here) and that there is a *unique morphism* Φ' (of $\mathcal{C}$, not assumed a priori to be an isomorphism) such that

$$\pi_{k,n} \circ \Lambda_n \circ \Phi' = \lambda_{k,n} \circ p_k \quad \text{for all } k \in K, n \geq 0,$$

where $\pi_{k,n} : \prod_{k'} M_{k',n} \to M_{k,n}$ is the product projection; and that this Φ' is an isomorphism.

The explicit Φ above satisfies exactly this defining property: for $(x_k)_k \in \prod_k \lim_{\leftarrow n} M_{k,n}$ (so $x_k = (x_{k,n})_n$), both $\pi_{k,n}\big(\Lambda_n\big(\Phi\big((x_k)_k\big)\big)\big)$ and $\lambda_{k,n}\big(p_k\big((x_k)_k\big)\big)$ equal $x_{k,n}$. By uniqueness of the morphism with this property, $\Phi = \Phi'$; as Φ' is an isomorphism, so is Φ. ∎

Corollary 3.3.158 (Cofinal Reindexing of a Tower)
Let $\mathcal{C}$ be as in (3.3.154), $(M_n, \rho_n)_{n\geq 0}$ a tower in $\mathcal{C}$, and $n_0 < n_1 < n_2 < \cdots$ a strictly increasing sequence in $\mathbb{N}$. Then, writing $\tilde{\rho}_N := \rho_{n_{N+1},n_N} : M_{n_{N+1}} \to M_{n_N}$ for the composite restriction map of (3.3.155), $\left(M_{n_N}, \tilde{\rho}_N\right)_{N\geq 0}$ is again a tower in $\mathcal{C}$, and

$$(m_n)_{n\geq 0} \mapsto \left(m_{n_N}\right)_{N\geq 0}$$

defines an isomorphism $\lim_{\leftarrow n} M_n \xrightarrow{\sim} \lim_{\leftarrow N} M_{n_N}$ of $\mathcal{C}$.

Proof. Drafted by Claude Sonnet, reviewed by David Rufino.

Regard $I := \mathbb{N}$ as a directed preorder and, per (3.3.155), the tower $(M_n)_n$ as an inverse system $F : I^{\mathrm{op}} \to \mathcal{C}$ realising $\lim_{\leftarrow n} M_n$. Let $I' := \{n_N : N \geq 0\} \subset I$, a full subcategory: it is directed (a subset of the totally ordered set I), and cofinal, as $n_N \to \infty$ so every $n \in I$ satisfies $n \leq n_N$ for some N (2.6.127). The order-preserving bijection $\sigma : \mathbb{N} \to I'$, $\sigma(N) := n_N$, is an isomorphism of the directed preorders $\mathbb{N}$ and I', and $F\sigma : \mathbb{N}^{\mathrm{op}} \to \mathcal{C}$ is exactly the tower $\left(M_{n_N}, \tilde{\rho}_N\right)_N$ (its restriction map for $N < N+1$ is $F(\sigma(N) \to \sigma(N+1)) = \rho_{n_{N+1},n_N} = \tilde{\rho}_N$, and composites thereof match for $N < N'$); so, by (2.6.97) (applied to σ) together with (3.3.155) applied to the tower $\left(M_{n_N}, \tilde{\rho}_N\right)_N$, $\left(\lim_{\leftarrow n} M_n, \{\lambda_{n_N}\}_{N\geq 0}\right)$ realises $\lim_{\leftarrow i\in I'} F(i)$, and $\left(\lim_{\leftarrow N} M_{n_N}, \{\lambda'_N\}_{N\geq 0}\right)$ (the latter's own projections) also realises it; identifying these two cones over $F|_{I'^{\mathrm{op}}}$ via the (unique, by terminality) comparison isomorphism between them, which sends $(m_n)_{n\geq 0} \in \lim_{\leftarrow n} M_n$ (viewed as a cone-point of the first) to the compatible sequence $\left(m_{n_N}\right)_{N\geq 0} \in \lim_{\leftarrow N} M_{n_N}$ agreeing with it on projections $\lambda_{n_N} = \lambda'_N \circ$ (this isomorphism).

By (2.6.132) (as I' is cofinal in I), the canonical morphism of cones $\lim_{\leftarrow i\in I} F(i) \to \lim_{\leftarrow i\in I'} F(i)$ is an isomorphism, and by construction (restriction of the limit cone's own projections λ_n to $i = n_N \in I'$) it sends $(m_n)_{n\geq 0} \to \left(m_{n_N}\right)_{N\geq 0}$ under the identification of the previous paragraph. Composing the two isomorphisms of this and the previous paragraph gives the stated isomorphism $\lim_{\leftarrow n} M_n \xrightarrow{\sim} \lim_{\leftarrow N} M_{n_N}$, $(m_n)_n \to \left(m_{n_N}\right)_N$. ∎

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